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# Forecasting Project

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January 6, 2026



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2025–2026

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# 1 Introduction

This project studies the time-series behaviour and forecasting of the 10-year sovereign yield spread between Spain and Germany, a key factor of fiscal solvency, market risk and financial fragmentation within the euro area. Within a monetary union that shares a common currency and monetary policy, such spreads condense information about country-specific fundamentals as well as market perceptions of underlying risk. Understanding and forecasting the Spain–Germany 10-year spread is therefore central for assessing the financial stability of European countries. We select Spain as our "peripheral" country and Germany as the "core" issuer, as German Bunds are considered the risk-free asset in the EU.

The report combines a theoretical and empirical perspective, in line with the forecasting course taught by Professor Candelon, by linking macro-finance theory of the term structure with modern time-series methods. The first part develops the economic theory of yield spreads.

Within this framework, the section identifies a set of theoretically grounded causal variables such as growth, HICP inflation, unemployment, current-account and fiscal indicators, sovereign CDS, financial volatility and political uncertainty that will later enter the forecasting models as key drivers of the Spain–Germany spread.

The empirical analysis then proceeds in three stages. First, a univariate Box–Jenkins approach is used to estimate ARIMA models for the 10-year spread and to generate forecasts up to three years ahead, with out-of-sample performance evaluated separately for the pre- and post-COVID periods. Second, multivariate models (VAR-X and VECM-X) are estimated to capture the dynamic interactions between the spread and our carefully selected exogenous variables, again producing three-year-ahead forecasts and comparing their accuracy across sub-periods. Third, an innovative component, based on the Kalman-filter state-space model is developed, providing an alternative representation of the spread's dynamics and allowing for richer real-time forecasting and structural interpretation.

Throughout the project, the data are drawn from high-quality sources, the Federal Reserve Economic Data (FRED) database, the European Central Bank, Eurostat, Investing.com and the Bloomberg terminal, ensuring consistency and reliability of the time series used in the analysis. By combining rigorous theory, careful variable selection and a range of forecasting methodologies, the study aims to deliver both accurate forecasts of the Spain–Germany 10-year yield spread and deeper insight into the economic forces that drive it, offering useful evidence for policymakers, market participants and researchers interested in euro-area sovereign-risk dynamics.

## 2 Theoretical analysis of the yield spread

This theoretical discussion prepares the ground for an empirical section in which the yield spread between Spain and Germany will be modeled and forecast as a function of macro-financial fundamentals and risk indicators. The aim of the theory section is therefore twofold: to present the main economic theories of yield spreads and their information content, and to identify, already at this stage, which variables are plausible causal drivers of the Spain–Germany 10-year spread.

### 2.1 Bond yields and the expectation hypothesis

In this section, we understand bond yields and their existing types. We also review economic theories such as expectation hypothesis and risk premia.

In bond markets, the yield is the annualized return an investor expects to earn by holding a bond to maturity, given its price, coupon payments and redemption value.

Under the expectations hypothesis, the yield on an  $n$ -period bond is roughly equal to the average of current and expected future short-term interest rates over the life of the bond (Crump et al., 2016). We can write the following equation:

$$y_{n,t} \approx \frac{1}{n} \sum_{k=0}^{n-1} \mathbb{E}_t r_{t+k}$$

$r_{t+k}$  is the short-term risk-free rate  $k$  periods ahead.

The expectations hypothesis fails dramatically at explaining the behavior of bond yields. While this theory is applicable for short-term yields, it cannot be applied to bonds with medium to long term maturities.

The term-structure theory considers that the yield is the value computed under the expectation hypothesis and a risk premium. This term premium is added to compensate investors for bearing interest-rate, liquidity and credit risk over longer horizons. Term premiums actually account for most of the variations in long term yield, and it gains importance when maturity is further away. For bonds with a maturity over 3 years, term premiums are the main drivers (Crump et al., 2016).

Yields are analyzed along two key dimensions, maturity and issuer. We differentiate the type of bonds by maturity. There exists short (under 2 years), medium (2 to 7 years) and long-term bonds (10 years or over). Depending on their longevity, they have different exposures to expectations and risk premia. They are also different depending on the issuer. We distinguish very safe sovereign issuers such as Germany from riskier sovereigns, banks or corporations whose bonds embed higher risk premia and thus have to offer higher yields for a given maturity. Within the euro area, German Bunds are treated as the closest available benchmark, so other counties' yields, such as Spanish yields, are viewed as carrying additional risk premia and it is reflected by a positive yield spread between Spain and Germany.

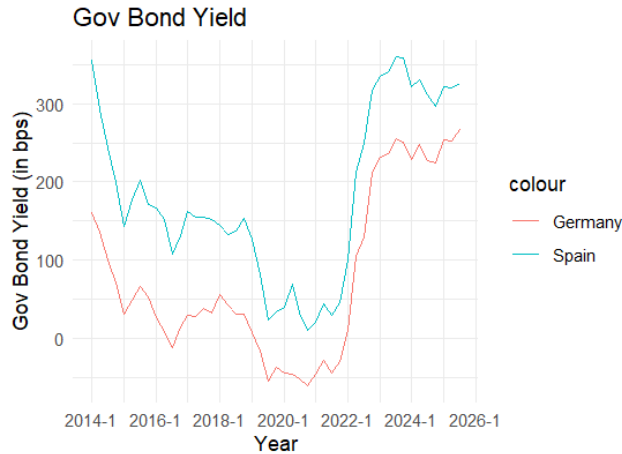


Figure 1: Spanish and German Bond Yields (2014-2025)

This figure describes the German and Spanish bond yields, and is consistent with theory. German debt is perceived as safe and investors demand lower yields.

## 2.2 Theory over yield spreads

A yield spread is the difference between two yields, and its meaning depends on how the two instruments are chosen. One possibility is to calculate the term spread, defined for a given issuer as the difference between a long-term and a short-term yield, which captures the slope of the yield curve and, in term-structure theory, reveals information about expectations of future short-term interest rates and about the term premium.

$$TS_t = y_{LT,t} - y_{ST,t}$$

Another central category is the sovereign spread at a given maturity, obtained by subtracting the yield on a benchmark sovereign from the yield on a riskier sovereign of the same maturity. In the euro area, it is usually constructed as the yield of a European country minus the yield of the German Bund, so that the spread isolates compensation for country-specific credit, liquidity and other risks relative to Germany.

In this project we will not analyze the term spread. Our focus is on the 10-year sovereign yield spread of Spain relative to Germany, defined as

$$SpreadBY_t^{ES-DE} = y_{10,t}^{ES} - y_{10,t}^{DE}.$$

This time series can be interpreted as the extra yield investors require to hold Spanish 10-year bonds instead of 10-year German bonds, given that both are denominated in the same currency and share the same common monetary policy. This extra yield compensates for perceived differences in default risk, liquidity, and other risk components for example a redenomination risk that became important during the euro-area crisis.

While term spread are a commonly used indicator of monetary policy and future economic outlook, sovereign spreads provide mostly information about relative creditworthiness, fiscal and external vulnerabilities, and market stress. In the euro area, widening sovereign spreads are interpreted as signals that markets perceive a higher country risk, as explained above (default, restructuring, liquidity or redenomination risk) for the affected sovereign. Episodes of rapid spread widening, especially when accompanied by rising sovereign CDS and global volatility, are usually seen as manifestations of financial stress and flight-to-quality into German Bunds, whereas periods of spread compression can be read as evidence that risk premia are declining and confidence is being restored.

## 2.3 Yield spreads within a monetary union

Within the European monetary union, member states share a common currency and a common monetary policy, so differences in short-term risk-free rates are largely eliminated. Cross-country differences in long-term yields therefore reflect mainly country-specific risk premia around a shared risk-free curve, making sovereign spreads particularly informative about relative fundamentals and risk perceptions. For Spain and Germany, the 10-year spread captures how much additional compensation investors demand to hold Spanish debt rather than German debt, given that both are denominated in euros and subject to the same ECB policy rate (Afonso et al., 2015; Crump et al., 2016).

As expected, the spread between the Spanish and German 10Y bond yields is constantly above 0. Investors perceive The former as a riskier sovereign than the latter, thus demanding a higher yield.

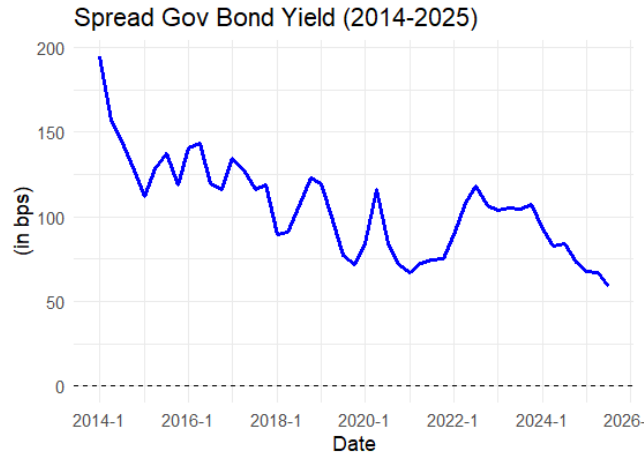


Figure 2: Spread of ES-DE Sovereign 10Y Bond Yields.

Empirical work on euro-area sovereign spreads shows that, in the first years of monetary union, spreads were relatively low and largely driven by common global and euro-area risk factors, with weak differentiation across countries based on fiscal fundamentals. During and after the global financial crisis, and especially in the sovereign debt crisis, investors became much more sensitive to country-specific fundamentals such as debt levels, deficits, growth prospects, and banking-sector vulnerabilities, and investors also began to price redenomination risk, that is, the possibility that some member states might exit the monetary union or restructure in a different currency. Institutional developments such as the creation of the European Stability Mechanism and the ECB's non-standard measures, including large-scale asset purchases and conditional backstop programmes, then altered the pricing of these risks by reducing tail scenarios and compressing spreads.

## 2.4 The euro-area sovereign debt crisis and Spain–Germany spreads

The euro-area sovereign debt crisis was a prolonged period, between 2010 and 2012 of acute financial stress during which several European governments, in particular Greece, Ireland, Portugal, Spain, and Italy experienced sharp increases in borrowing costs and widening bond yield spreads relative to Germany. These developments raised serious concerns about their ability to remain solvent within the monetary union. While the crisis followed the 2008 global financial downturn, it exposed structural weaknesses specific to Europe, including a monetary union without a fully integrated fiscal framework, significant pre-crisis imbalances between core and peripheral economies, and strong interdependencies between national banking systems and sovereign balance sheets (De Santis, 2012; Wikipedia contributors, 2025)

A key driver of the crisis was the accumulation of both private and public debt during a prolonged period of low interest rates following the introduction of the euro. Financial markets largely priced sovereign debt across member states as similarly safe, despite materially different economic fundamentals. Peripheral economies such as Spain, Ireland, and Greece benefited from what was effectively an implicit guarantee associated with Germany, gaining access to low-cost financing. This environment fuelled housing booms, rapid credit expansion, and, in some cases, sustained fiscal imbalances and accounting practices that obscured the true condition of public finances. When the global financial crisis struck, asset prices fell sharply, capital flows reversed, and governments were compelled either to stabilise their banking systems, as in Ireland and Spain, or to address suddenly unsustainable fiscal positions, as in Greece. As a result, a crisis that initially centred on banks and external balances evolved into a full-scale sovereign debt crisis.

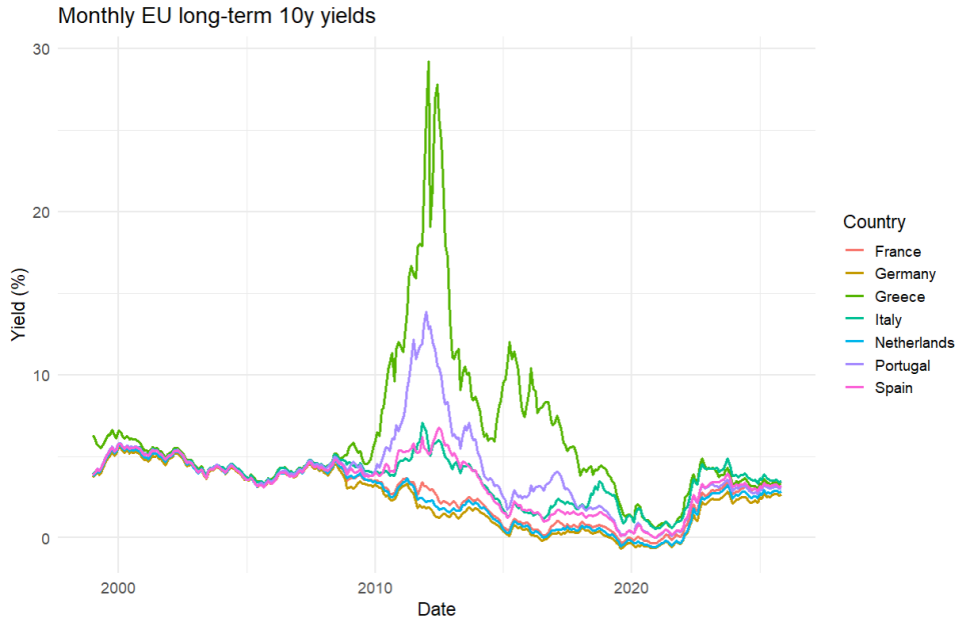


Figure 3: Sovereign yields of EU countries

The crisis also highlighted significant trade and current-account imbalances within the euro area. Core economies, particularly Germany, recorded large and persistent current-account surpluses, while many peripheral countries accumulated substantial deficits financed by capital inflows from the core. Because euro-area members share a common currency, deficit countries could not rely on exchange-rate devaluation to restore competitiveness. Instead, adjustment occurred through internal devaluation, including wage moderation, price compression, recessionary conditions, and fiscal consolidation. These measures weakened economic growth and worsened debt dynamics. As concerns over solvency and competitiveness intensified, sovereign bond spreads relative to German Bunds widened significantly, and fears of financial contagion grew due to the extensive exposure of core-country banks to peripheral sovereign debt.

In Spain's case, the sovereign debt crisis was closely linked to the collapse of a large property and construction boom financed by cross-border bank lending and wholesale funding. When the housing market downturn materialised, Spanish banks incurred substantial losses, prompting government intervention and participation in a euro-area bank recapitalisation programme. This support led to a rapid increase in public debt and fiscal deficits from previously moderate levels. Investor confidence deteriorated, with rising doubts about the sustainability of Spain's public finances and the robustness of its banking sector. Consequently, Spanish 10-year government bond yields rose sharply, and the spread over German Bunds exceeded 500–600 basis points at the peak of the crisis, reflecting not only heightened default risk but also a significant redenomination risk premium driven by fears of a potential exit from the euro area.

## 2.5 Causal variables for the Spain–Germany 10-year spread

We selected causal variables following the term-structure and risk-premium theories as well as the euro-area sovereign-spread literature. It is evident that the main factors of spread variation are growth, inflation, job market conditions, external balances, fiscal fundamentals, credit-risk indicators, global volatility and political uncertainty. Therefore, these are our selected variables:

- real GDP growth,
- Harmonized Index of Consumer prices (HICP) inflation,

- Unemployment rate,
- debt-to-GDP ratio,
- Budget Balance (net lending/borrowing, % of GDP),
- credit default swaps
- Trade Balance of commodities (% of GDP).

For all the variables cited above, we take the difference between the Spanish and German levels, to get these time series as spreads.

- VSTOXX volatility index

This variable is expressed in levels and its values are in percentages.

The **spread of real GDP growth** is defined as :

$$\Delta g_t = g_t^{ES} - g_t^{DE},$$

where  $g_t^{ES}$  and  $g_t^{DE}$  denote the real GDP growth rates of Spain and Germany. Real GDP growth is the percentage change in a country's output of goods and services after adjusting for inflation. Stronger growth in Spain relative to Germany improves Spain's fiscal outlook by enlarging the tax base and reducing unemployment-related spending, thus lowering perceived default risk and the required credit-risk premium. From theory, we expect Germany to have higher growth therefore we would mostly get negative  $\Delta g_t$  values. In the same logic, a higher  $\Delta g_t$  should be associated with a lower Spain–Germany spread.

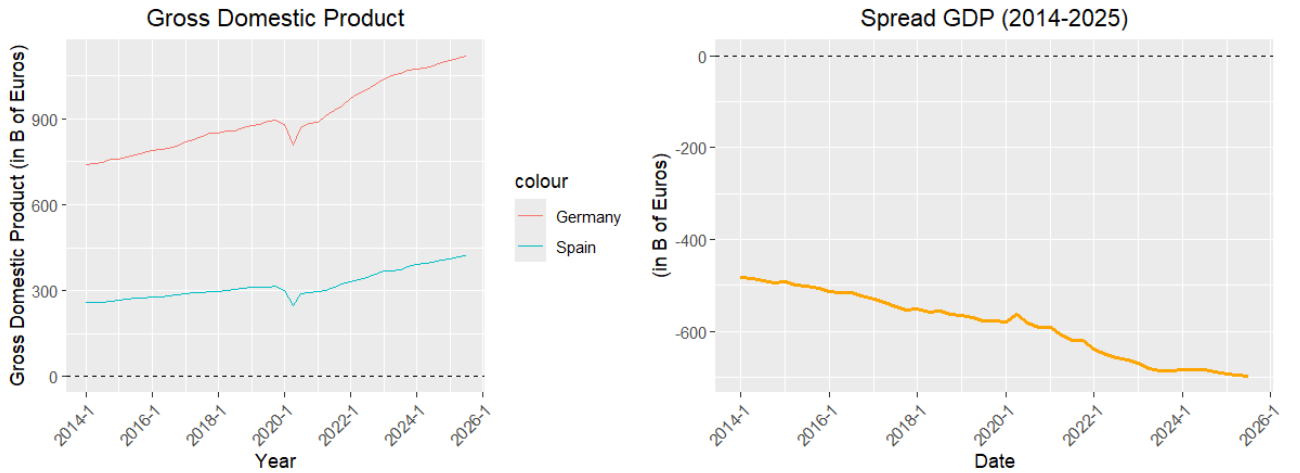


Figure 4: GDP levels and spread

We observe that the Spread in GDP (in Billions of Euros) has steadily moved further into negative territory, going from -480B to nearly -700B. This indicates that Germany's economy grew faster than Spain's, making the gap in total size bigger.

The **spread of HICP inflation** is defined as:

$$\Delta \pi_t = \pi_t^{ES} - \pi_t^{DE},$$

with  $\pi_t^{ES}$  and  $\pi_t^{DE}$  measuring inflation. HICP is an index of prices, not directly an inflation rate. It measures the cost of a fixed basket of consumer goods and services. Inflation is defined as the rate of change of this index over time. To get the growth rate of each series, they

are transformed in log-differences. Regarding possible spread values, we expect inflation of a stable country to have a constant mean. Persistent inflation differentials would signal losses of competitiveness and a need for future internal devaluation. Theoretically, large positive inflation differentials correlate with higher sovereign spreads, which is consistent with markets penalizing exchange-rate misalignments.

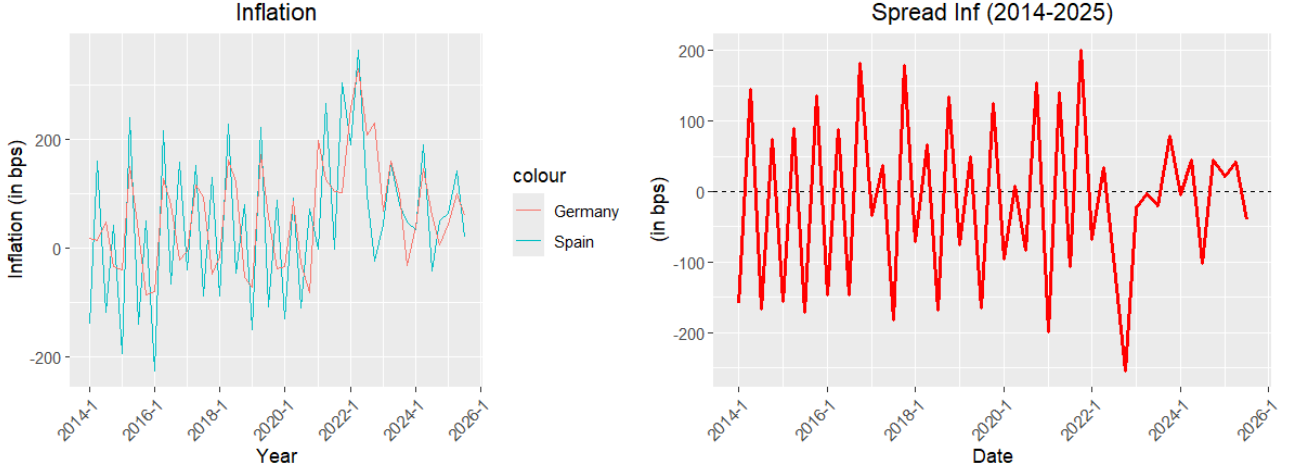


Figure 5: Inflation levels and spread

As we can observe, the spread in Inflation has lots of ups and downs, moving between -2% and +2% without a clear long-term direction. This shows that price differences are mostly just short-term changes, and there don't seem to be the need for a Spanish monetary devaluation.

The spread of the **unemployment rate** is given by

$$\Delta u_t = u_t^{ES} - u_t^{DE},$$

where  $u_t^{ES}$  and  $u_t^{DE}$  denote Spanish and German unemployment rates. Higher unemployment in Spain relative to Germany indicates weaker economic conditions and lower potential output, which tend to depress tax revenues, increase social transfers and heighten concerns about social and political stability. From a credit-risk perspective, higher  $\Delta u_t$  should therefore increase the Spain-Germany spread. We expect  $\Delta u_t$  values to be mostly positive.

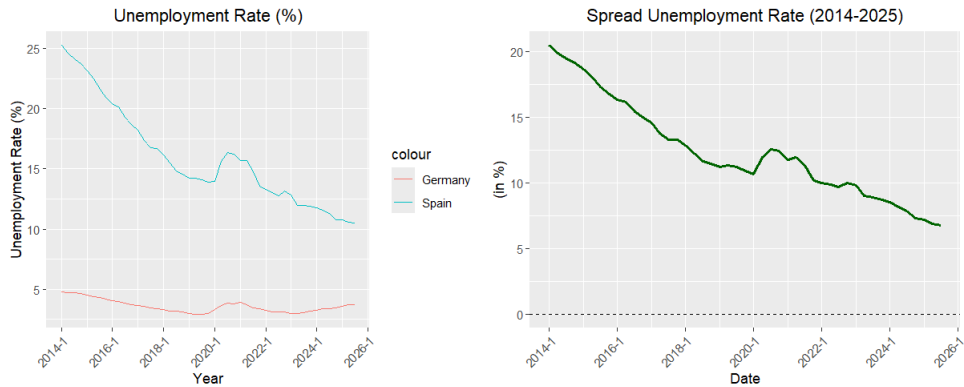


Figure 6: Unemployment rates levels and spread

Labor market numbers show a very strong convergence. The Spread in the Unemployment Rate (in %) started at a huge 20% and steadily went down to around 7% by mid-2025. This suggests a long and deep reduction of the labor gap between the two countries.

The spread of the **public debt-to-GDP ratio** is

$$\Delta d_t = d_t^{ES} - d_t^{DE},$$

with  $d_t^{ES}$  and  $d_t^{DE}$  denoting gross public debt-to-GDP in Spain and Germany. Higher relative debt in Spain directly raises concerns about fiscal sustainability and default risk and is one of the most robust determinants of euro-area sovereign spreads in empirical work, often with non-linear effects at high debt levels. We therefore predict a positive relationship, with the possibility that the marginal impact of debt on spreads increases once debt exceeds specific thresholds.

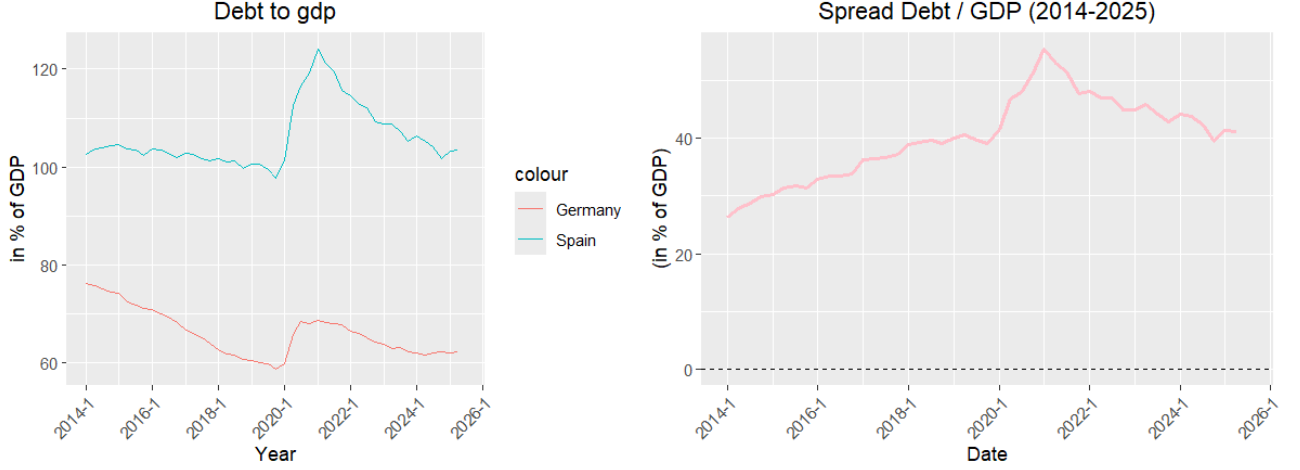


Figure 7: Debt to GDP levels and spread

The Spread in Debt-to-GDP followed a rising trend, peaking at 55% during the 2020 crisis before going down to 40% in 2025. As explained above, this increase in relative debt is likely to have a steepening impact on the yield spread.

The spread of the **budget balance** (net lending/borrowing, as a percentage of GDP) is defined as

$$\Delta BB_t = BB_t^{ES} - BB_t^{DE},$$

$BB_t$  is the difference between government revenues and expenditures where a positive value denotes a surplus and a negative value a deficit. In this equation, a negative  $\Delta BB_t$ , means a larger deficit in Spain than in Germany, and signals weaker fiscal discipline and a deteriorating debt trajectory, which should raise the credit-risk premium embedded in Spanish yields relative to German yields. The spread of this variable has therefore a flattening impact on the yield spread when it lowers, and steepens the curve when the value decreases.

the Spread in the Budget Balance showed a steady increase, rising from a gap of -7% in 2014 to almost -1% in 2025. This provides a positive outlook for Spain, as they seem to have a better control of the budget compared to Germany over the long run.

The spread in sovereign **credit default swaps** (CDS) is

$$\Delta CDS_t = CDS_t^{ES} - CDS_t^{DE},$$

A sovereign CDS is an insurance-like contract that pays out if a government defaults or re-structures its debt. In our case, we take 5-year sovereign CDS spreads for Spain and Germany. Sovereign CDS directly price market assessments of default probability and loss given default

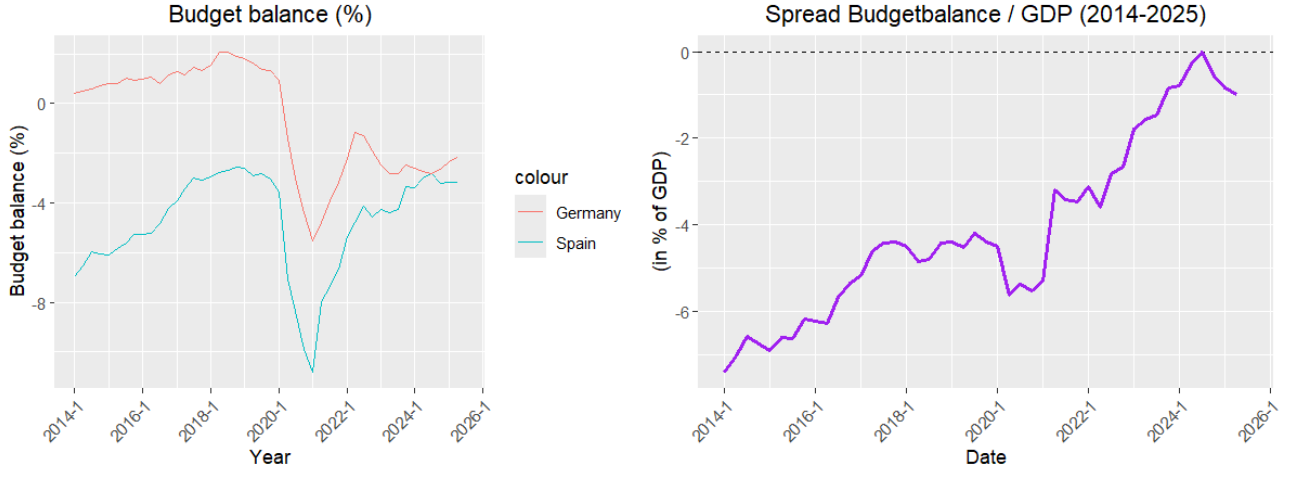


Figure 8: Budget balance levels and spread

and often respond more quickly than bond yields to news about fundamentals or risk sentiment. A higher Spanish CDS relative to German CDS reflects increased perceived default risk for Spain. Thus, we expect CDS and yield spreads to be highly correlated and causal.

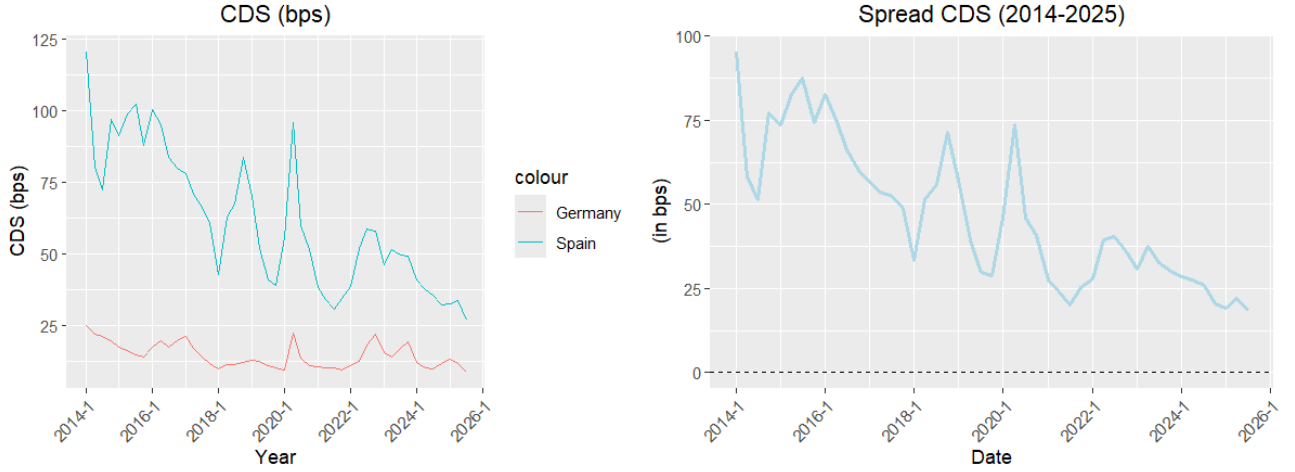


Figure 9: CDS levels and spread

The Spread in CDS dropped from nearly 100 bps to about 20 bps, with a short jump during the 2020 COVID-19 shock. This shows a big decrease in the perceived risk of Spain compared to Germany.

The spread of the **trade balance of commodities** as a percentage of GDP, can be defined as

$$\Delta TB_t = TB_t^{ES} - TB_t^{DE}.$$

The trade balance of commodities measures the difference between exports and imports of goods, and they measured as a percentage of GDP. The spread compares Spain's goods trade balance with Germany's. A weaker Spanish trade balance relative to Germany suggests worse terms of trade and more dependence on external financing, which increases perceived external vulnerability and is expected to raise the Spain–Germany sovereign spread.

The Spread in Trade Balance for Commodities (in Billions of Euros) stayed at low levels, between -40B and -75B, with a large jump during 2020 when global trade was interrupted. Germany has a consistent trade surplus and Spain a deficit. This makes the latter more dependent on other countries and vulnerable to externalities.

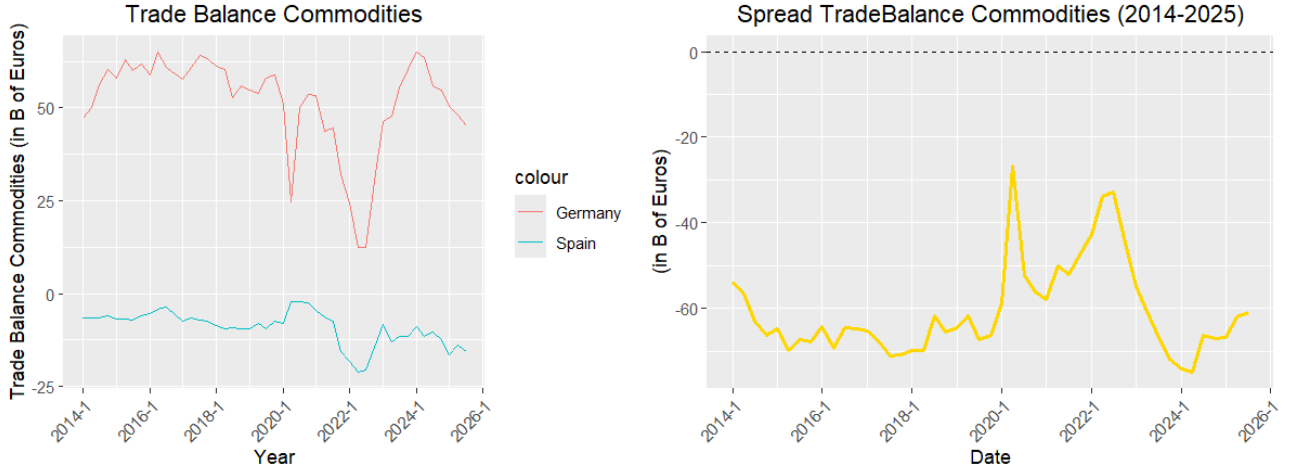


Figure 10: Trade Balance of commodities levels and spread

### VSTOXX (european equity volatility index)

VSTOXX is an implied-volatility index based on a combination of call and put options on the Euro Stoxx 50 equity index. It is equivalent to the american VIX and is often described as Europe’s “fear gauge” because it rises when expected stock-market volatility and risk aversion increase. It is used as a common euro-area risk factor: when VSTOXX is high, investors are more risk-averse and tend to demand higher premia on risky sovereigns like Spain while seeking safety in German Bunds, widening the Spain–Germany spread.

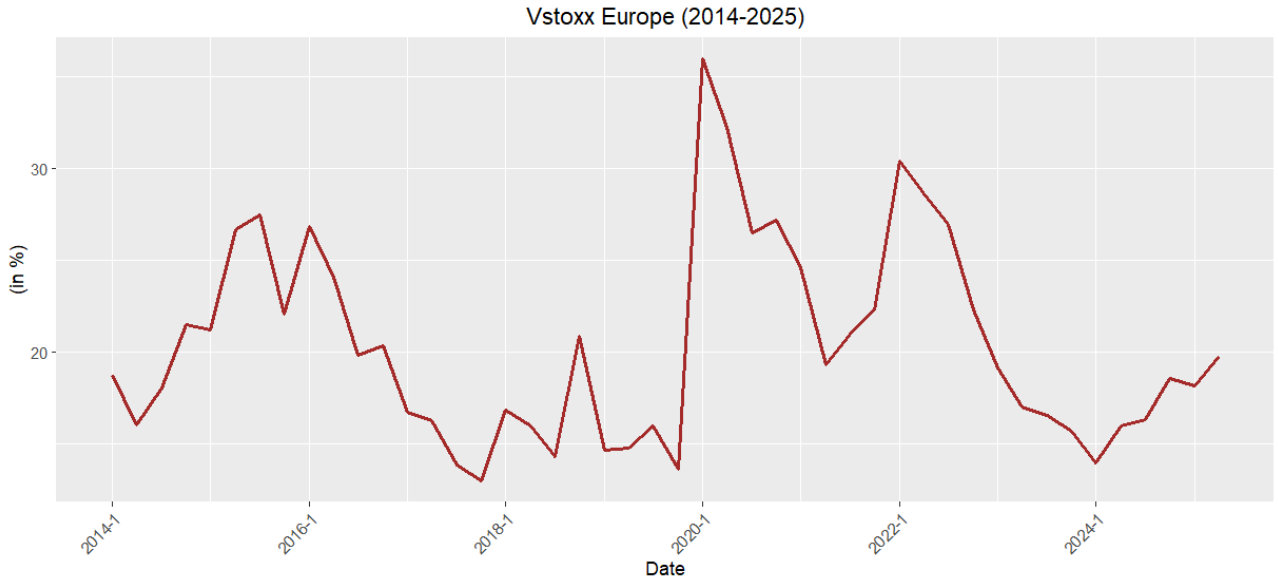


Figure 11: VSTOXX levels

Taken together, these variables map closely into the theoretical channels identified earlier. Growth, inflation and unemployment capture the macroeconomic outlook and influence expectations about future fiscal capacity; the current account and trade balance of commodities measure external sustainability; debt and budget balances represent fiscal solvency; CDS and VSTOXX capture market-based and political risk premia. Specifying them mainly as differentials between Spain and Germany aligns the regressors conceptually with the dependent variable  $S_t^{ES-DE}$  and embodies the idea that the Spain–Germany 10-year sovereign spread is an equilibrium outcome of how markets price Spanish risk relative to the German benchmark within a common monetary union.

## 3 Univariate forecast

### 3.1 Theory and common thread

Before we talk about the results of our ARIMA model, it is important to explain what it is and how the forecasting process works. The ARIMA model, which stands for Auto Regressive Integrated Moving Average, is one of the most common tools for predicting time series. It is used for univariate data, meaning we try to guess future values based only on the past of that specific and only variable. The model combines three main parts: AR, I, and MA to understand different patterns in the data.

The Auto Regressive (AR) part looks at the relationship between the current value and a certain number of past values, which are called lags. For example, in an AR(1) process, the value today depends on the value from yesterday, multiplied by a coefficient that the model calculates. This part is very useful for finding dependencies over time.

The integrated (I) component is used to make the series stationary, which is a key assumption for ARIMA modelling. Stationarity means that the statistical properties of the series, such as its mean and variance, do not change over time, and that its autocorrelation structure depends only on the lag and not on time. For non-stationary series, such as those exhibiting trends, the integration step consists of differencing the data one or several times in order to eliminate trends and other sources of non-stationarity.

The Moving Average (MA) part focuses on the link between the current value and past forecast errors. This helps the model account for random shocks or unexpected changes, which helps smooth out the noise and makes the predictions better. We describe an ARIMA model using three parameters:  $p$ ,  $d$ , and  $q$ . Here,  $p$  is the number of lags for the AR part,  $d$  is the number of times we differenced the data, and  $q$  is the number of past errors used in the MA part.

The process of building an ARIMA model usually follows a few steps. First, we have model identification. This is where we choose the right values for  $p$ ,  $d$ , and  $q$ . We can use plots like the ACF and PACF to help us decide, or we can use an automatic tool like the `auto.arima` function in R.

Second, we have model fitting. We use our historical data to estimate the parameters. The goal here is to minimize the residuals, which are the differences between the real values and the ones predicted by the model.

Third, we do the diagnostics. We analyze the residuals to make sure the model captured all the patterns in the data. We want the residuals to look like white noise. To check this, we use the Breusch-Godfrey test to make sure there is no autocorrelation. We also use the Shapiro-Wilk test to see if the residuals follow a normal distribution. Finally, we use the Breusch-Pagan test to check for heteroscedasticity, which helps us see if the variance of the errors is constant.

Finally, we move to forecasting. Once the model is checked and validated, we use it to predict future values. We usually show these forecasts with 80% and 95% confidence intervals to show the level of uncertainty in the predictions.

### 3.2 Arima activation

The objective of this part is to forecast the future values of the first difference of the government bond yield spread based on historical quarterly data, allowing for a better understanding of how market risk and financial pressure might change over time. This analysis aims to provide useful insights into future market behavior and potential shocks. The data is structured

as a univariate time series, which means it consists of a single variable, the quarterly change in the spread, tracked over a consistent time interval. Using a quarterly frequency allows us to capture important economic dynamics and patterns while avoiding the extreme noise and complexity usually associated with daily or monthly variations. This approach is particularly helpful when the main goal is to identify the general direction of the spread and assess how broader economic factors influence it, rather than focusing on very short-term movements. By using this structured dataset, the analysis aims to find meaningful relationships in the past data, which can then be used to generate realistic predictions for the upcoming periods.

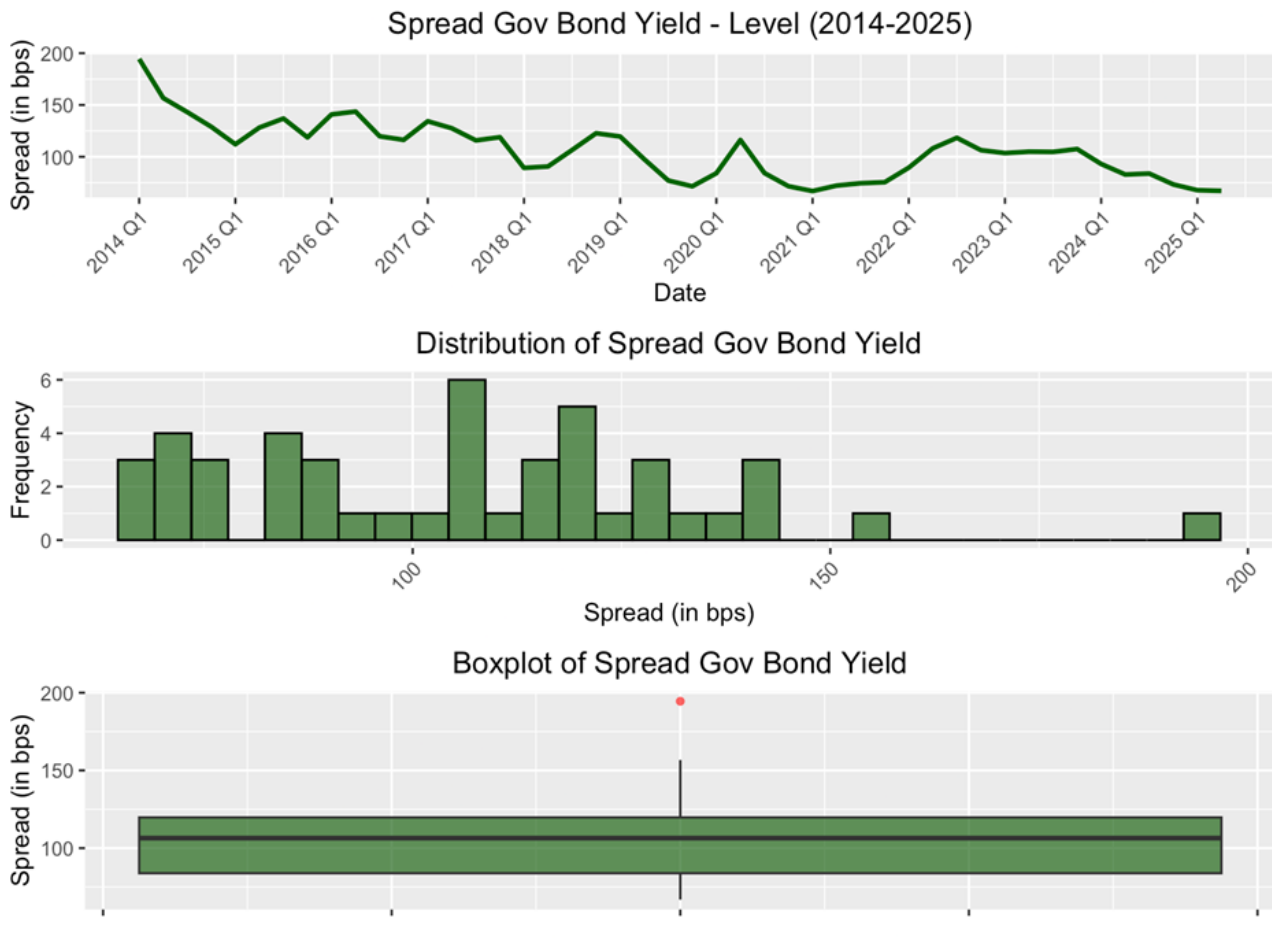


Figure 12: Analysis of the Spread Gov Bond Yield

The exploratory data analysis of the Spread Gov Bond Yield in levels shows how the spread between the two countries changed between 2014 and 2025. The time series plot reveals that the spread started very high, near 200 bps in early 2014, before dropping quickly. After this initial fall, the values fluctuated between 75 and 150 bps with some small peaks in 2019 and 2022. Overall, the trend is moving downwards, and the spread reached its lowest points at the end of the period in 2025.

The histogram gives more details about the distribution of these values. Most of the data is concentrated between 75 and 130 bps, showing that the spread stayed in this range for most of the time. The distribution is not symmetric and shows a long tail stretching to the right because of the few high values. There is a clear gap between the main group of data and the higher values, with one isolated bar near 200 bps. This suggests that the very high value at the start of the series is an outlier compared to the rest of the observations.

Finally, the boxplot confirms the presence of this outlier in the data. The median is located around 110 bps, and the central box shows that half of the observations are between 85 and

120 bps. The interquartile range (IQR), which captures the central bulk of the data, shows that the spread stayed relatively consistent within this range for most of the quarters. The upper whisker reaches about 160 bps, but there is a specific red point far above it near 200 bps. This point represents the outlier we saw at the beginning of the time series, confirming that the early period was an exceptional phase with much higher risk.

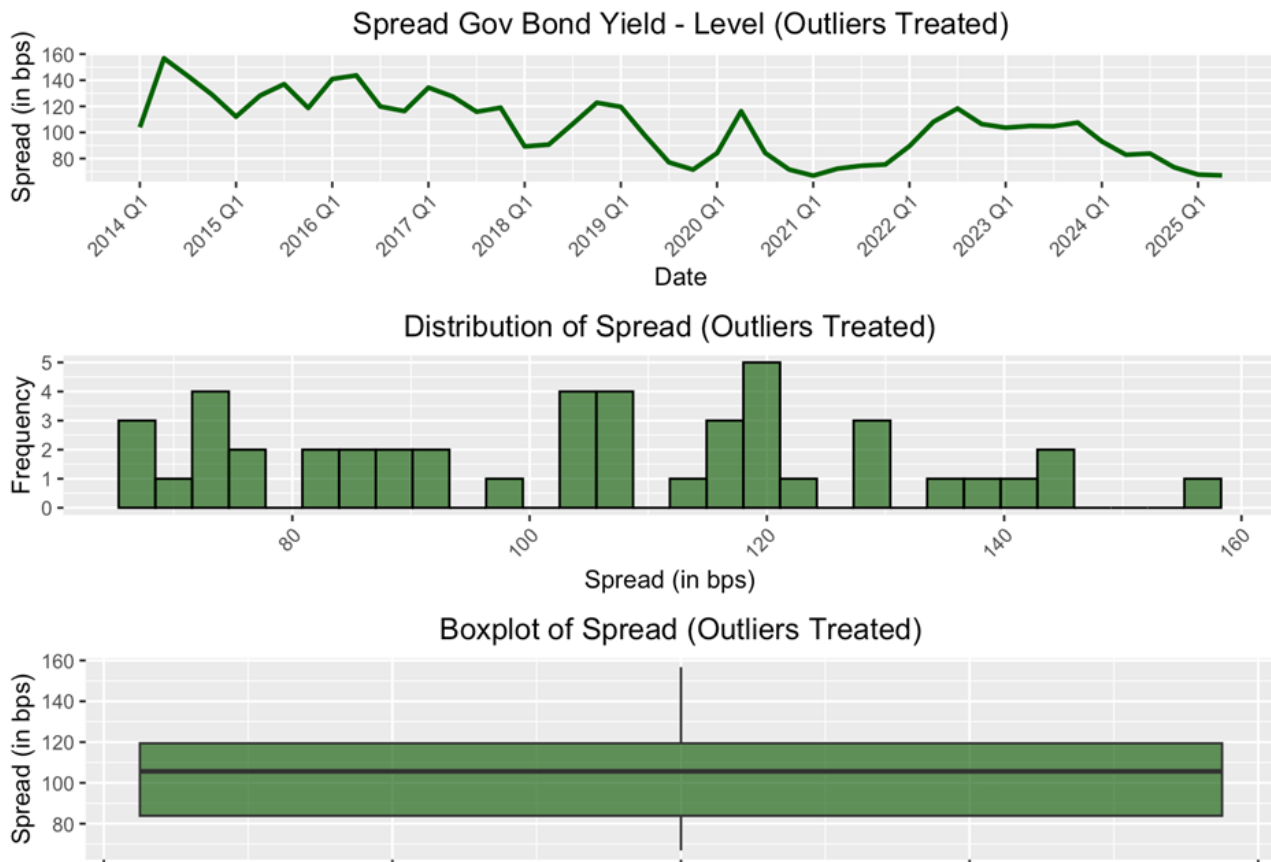


Figure 13: Analysis of Gov bond yields with outliers removed

Using the IQR test, we effectively found the only outlier of Q1 2014. We replaced him with the average value of the spread gov bond yield. Here in the figure above are the same plots but without outliers.

Now we can begin to check for stationarity of the time series. It is impossible to forecast a time series that is not stationary with an ARIMA model. If the series is non-stationary, we have to differentiate it to make it stable. A stationary time series is one where statistical properties like the mean, variance, and autocorrelation do not change over time, meaning the behavior stays the same no matter when you observe it.

To check for this, we use the augmented Dickey-Fuller test. In this statistical test, the null hypothesis  $H_0$  is that the series has a unit root, which means it is not stationary. The alternative hypothesis  $H_1$  is that the series is stationary.

We use the `adf.test` function from the `tseries` package in R. If the result shows a p-value lower than 0.05, we can reject the null hypothesis and conclude that the series is stationary. This test is very useful because it accounts for autocorrelation by using lagged differences in its calculation.

Because the augmented Dickey-Fuller test showed that the spread in levels was not stationary, we calculated the first difference and performed the same exploratory data analysis on this new variable.

### Augmented Dickey-Fuller Test

```
data: df_SpreadGovBY_Level$SpreadGovBY_Level
Dickey-Fuller = -2.115, Lag order = 3, p-value = 0.5282
alternative hypothesis: stationary
```

Figure 14: ADF test on the spread

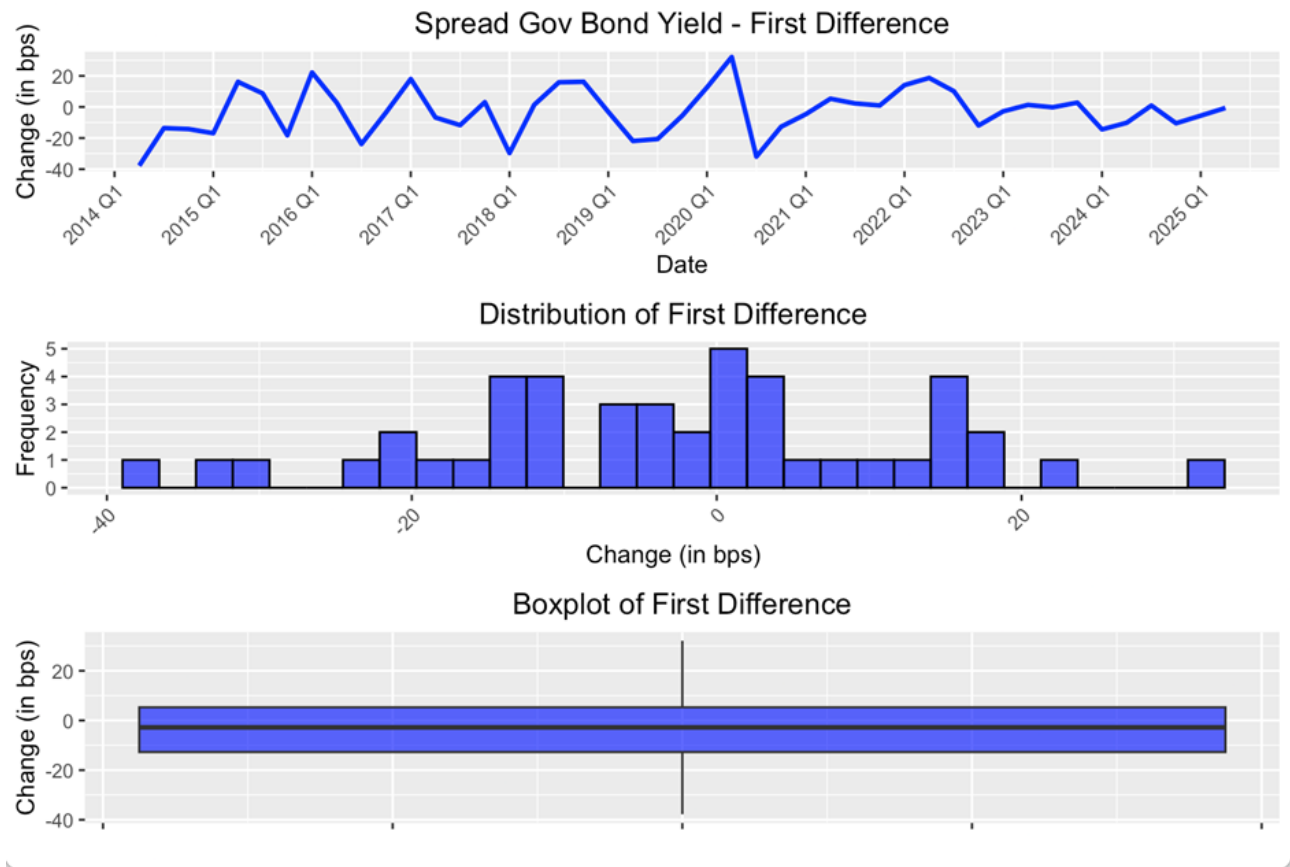


Figure 15: Analysis of the stationary spread time series

The exploratory data analysis of the Spread Gov Bond Yield in first differences this time shows how the spread variates between 2014 and 2025. The time series plot shows a lot of quarterly movement and volatility. It starts with a very sharp drop near -40 bps in early 2014. After that initial shock, the quarterly changes mostly fluctuate between -20 and 20 bps without a clear long-term trend. We can see another big peak in early 2020, where the spread increased quickly before falling back down right after.

The histogram gives us a better view of how these changes are distributed. Most of the values are centered around zero, which means that in many quarters, the change in the spread is relatively small. However, the distribution is quite wide and spread out across the horizontal axis. This confirms that the variable is volatile and frequently experiences significant ups and downs. The shape is more balanced than the levels plot.

The boxplot provides a final look at the stability of these changes. The median is located slightly below zero, suggesting that the spread had a small tendency to decrease on average during this period. The interquartile range (IQR), which captures the central bulk of the data, shows that half of the quarterly changes stayed between roughly -12 and 5 bps. While we do not see specific red dots for outliers in this version of the plot, the whiskers are very long,

especially on the negative side. This confirms that the massive drop at the very start was an extreme movement compared to the rest of the observed quarters.

Reusing one more time the IQR test, we effectively found no outlier at all. Hence the plots remain the same as before and we can begin to check for stationarity.

The augmented Dickey-Fuller test reveals to us that the p-value is 0.01 lower than 0.05 meaning that we reject  $H_0$  and the series is stationary. We now know that we don't need to differentiate the series anymore.

The next step is to look at the ACF and PACF plots of the variable to find the right values for  $p$  and  $q$ . The ACF of the variable measures the total correlation between the series and its past values at different lags to show the general pattern of the data. On the other hand, the PACF of the variable measures the partial correlation between the series and its past values at different lags (it only looks at the direct relationship with a specific lag, removing the influence of the periods in between). Using these two plots together helps us identify the correct order for our ARIMA model.

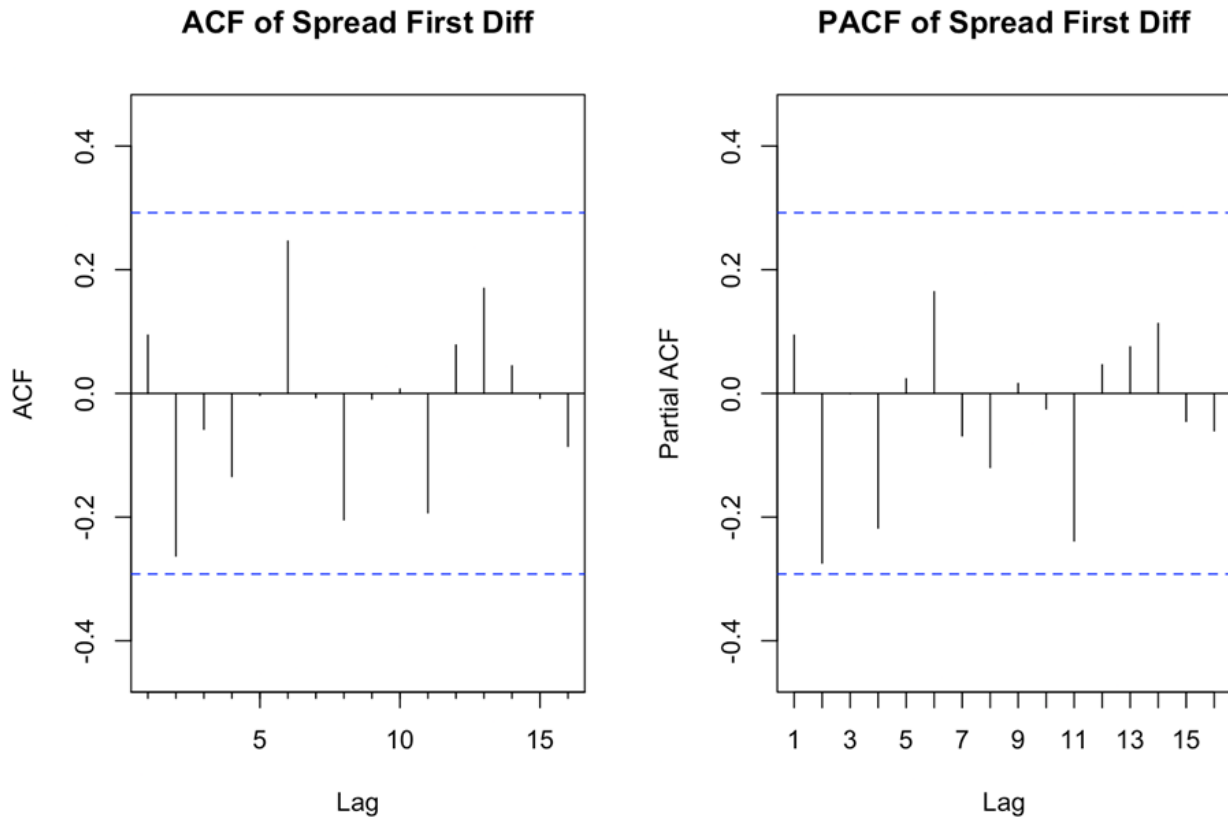


Figure 16: ACF and PACF plots of the stationary spread series

Based on the analysis of the ACF and PACF plots for the first difference of the spread, the determination of  $p$  (autoregressive order) and  $q$  (moving average order) for the ARIMA model is as follows. In the ACF plot, all the bars stay inside the blue dashed lines, which means there are no significant correlations. This indicates that there is no clear moving average (MA) component to include, as the quarterly changes already look quite random after differencing.

On the other hand, the PACF plot also shows that all the spikes are within the boundaries. There is no significant spike at the first lag or any other early lag, which suggests that the autoregressive (AR) part is not strongly needed. This pattern is very similar to white noise, where the past values do not have a direct and measurable impact on the current change,

meaning that the orders  $p$  and  $q$  should likely be zero.

Even though the ACF and PACF plots suggested an ARIMA(0,1,0) model, we decided to use the `auto.arima` function in R to find the ideal number of  $p$  and  $q$  parameters for our variable. This automatic tool tests several combinations to make sure we select the most efficient model for our forecasting. After running the function, the `auto.arima` function suggested an ARIMA(0,0,2) model with zero mean. It is important to note that the differentiation parameter  $d$  is zero because we ran the test directly on the variable that was already first differenced. This means our complete model is actually an ARIMA(0,1,2).

After implementing this ARIMA(0,0,2), looking at the model results is very helpful to see how well it works. By comparing the actual values (of our first diff spread) with the fitted values calculated by the ARIMA, we can better understand if the model captures the real movements of the spread.

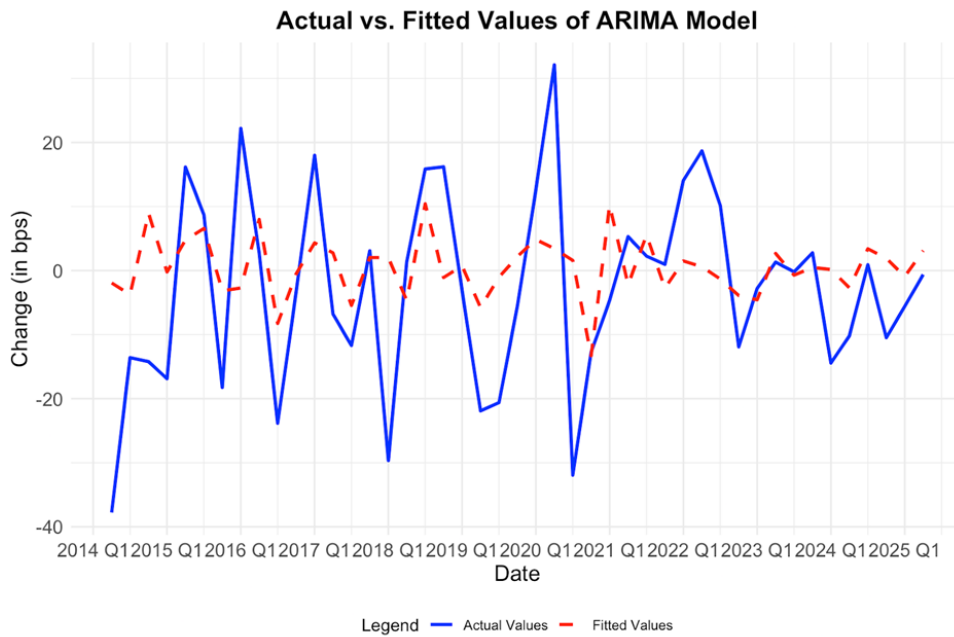


Figure 17: ARIMA model values compared to actual spread values

The plot shows the comparison between the actual values (blue solid line) and the fitted values (red dashed line) from the ARIMA model over time. Overall, the fitted values follow the general direction of the real data, but the model is much more stable and does not show the same level of intensity as the actual spread. It captures some of the basic patterns, but it stays closer to zero most of the time, which shows it has trouble following the very jumpy nature of the change of the spread.

While the model performs okay during periods with moderate changes, it misses the scale of the movements during times of high volatility. This is especially clear during the huge drop at the start of 2014 and the very sharp spike around early 2020. These differences show that the model is a bit too conservative and struggles to fully capture the sudden and extreme shocks that happened during this period.

### 3.3 Model evaluation through the global model, the pre-, and the post-covid periods

The provided table evaluates the ARIMA(0,0,2) model across three phases: the global model, the pre-covid data, and the post-covid data. We used several metrics to check the accuracy and

| Metric                                     | Model (Global) | Pre-covid | Post-covid |
|--------------------------------------------|----------------|-----------|------------|
| Shapiro-Wilk Normality (p-value)           | 0.8773         | 0.9973    | 0.7093     |
| Breusch-Pagan Heteroscedasticity (p-value) | 0.7207         | 0.8525    | 0.3438     |
| Breusch-Godfrey Autocorrelation (p-value)  | 0.9708         | 0.8821    | 0.9090     |
| MAE                                        | 11.1665        | 13.0940   | 9.1513     |
| RMSE                                       | 14.3993        | 15.9676   | 12.5520    |

Table 1: Model Performance and Diagnostic Metrics Comparison

the behavior of the residuals, including the Mean Absolute Error (MAE), the Root Mean Square Error (RMSE), and p-values from the Shapiro-Wilk, Breusch-Pagan, and Breusch-Godfrey tests.

The MAE measures the average magnitude of prediction errors. For the whole model, it is 11.1665. Before covid, it was 13.0940, but it dropped to 9.1513 after covid, suggesting better accuracy in the recent period. The RMSE, which penalizes larger errors more heavily than MAE, also decreased from 15.9676 pre-covid to 12.5520 post-covid, with a RMSE of 14.3993 for the entire Model. This shows a good improvement, but we must remember that the post-covid period is shorter of 1 observation, which can sometimes make the error values look a bit lower than they really are but in this case the difference of MAE and RMSE results between the 2 periods is too big.

All the statistical tests have p-values above 0.05, which means the model is valid. The Breusch-Godfrey test shows no autocorrelation in the residuals for any period, with p-values consistently above 0.88. The Shapiro-Wilk test confirms that the residuals are normal, especially pre-covid with a very high p-value of 0.9973. Finally, the Breusch-Pagan test shows that the variance is constant and stable for the three periods, although it is a bit less significant for the post-covid phase with a p-value of 0.3438. When comparing the two, the pre-covid period seems to be the best in terms of statistical reliability because the test results are much stronger. However, the post-covid period actually shows lower error values like MAE and RMSE. Overall, the model performs well and adapts correctly to both periods, which are based on datasets of comparable size. We analyzed the residuals of our global ARIMA(0,0,2) model and found that they remain well behaved both before and after the Covid-19 period. Splitting the sample shows that the Covid shock did not deteriorate the residual properties: post-2020 residuals remain homoskedastic, uncorrelated, and normally distributed, with even lower MAE and RMSE, indicating that the model is robust and reliable for future forecasting.

### 3.4 Forecasting

Following the estimation of the ARIMA model, we predicted the change of the spread government bond yield for Q3 2025 to Q3 2028. The results provide both the central point forecasts and their corresponding confidence intervals (80% and 90%) for each quarter of the year.

This forecast plot presents the historical data of the spread changes alongside the predicted values from Q3 2025 to Q3 2028. The black line represents the past quarterly variations, while the forecasted values are shown with a dashed blue line. The forecast indicates a relatively stable trend for the next three years, which is a big contrast to the strong fluctuations and peaks observed in the historical data. This suggests that the ARIMA model smooths out the past volatility to predict a more neutral path moving forward.

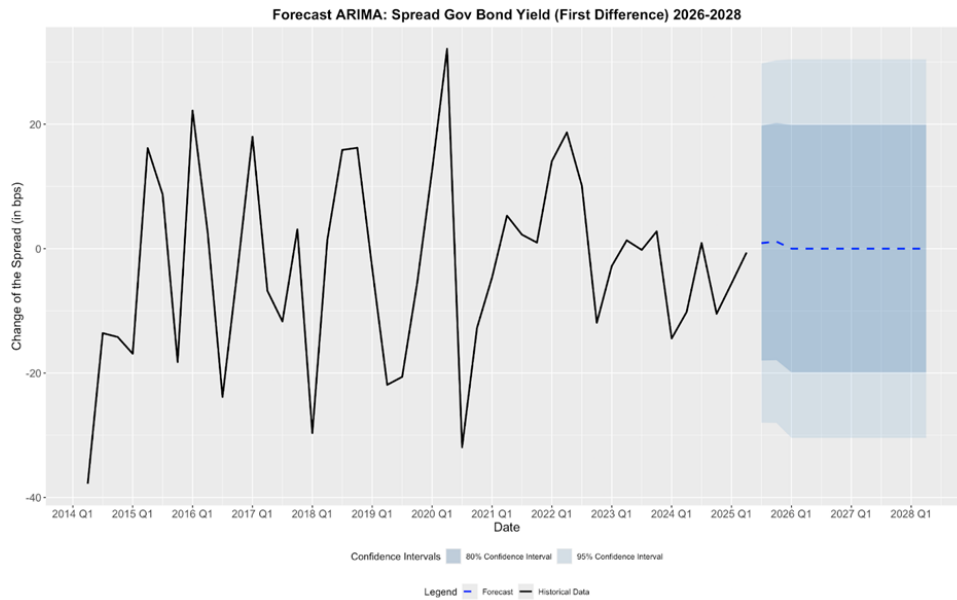


Figure 18: 1: ARIMA Forecast for the First Difference of Government Bond Yield Spread (2026–2028)

Looking closely at the start of the forecast, we can see a small rise in the spread change during the first quarters, starting in Q3 2025 and continuing in Q4 2025. After this initial little peak, there is a small descent starting in Q1 2026, and the line then stays almost perfectly flat at zero for the rest of the period until Q3 2028. This suggests that the model expects a very short-term adjustment before the quarterly changes completely settle down.

The shaded zones around the forecast represent the confidence intervals, with the darker blue showing the 80% level and the lighter blue showing the 95% level. We can see these zones widen as we move further into 2027 and 2028, showing that the model acknowledges higher uncertainty over time. This widening happens because it is much harder to be precise as the prediction moves further away from the last known data point in 2025.

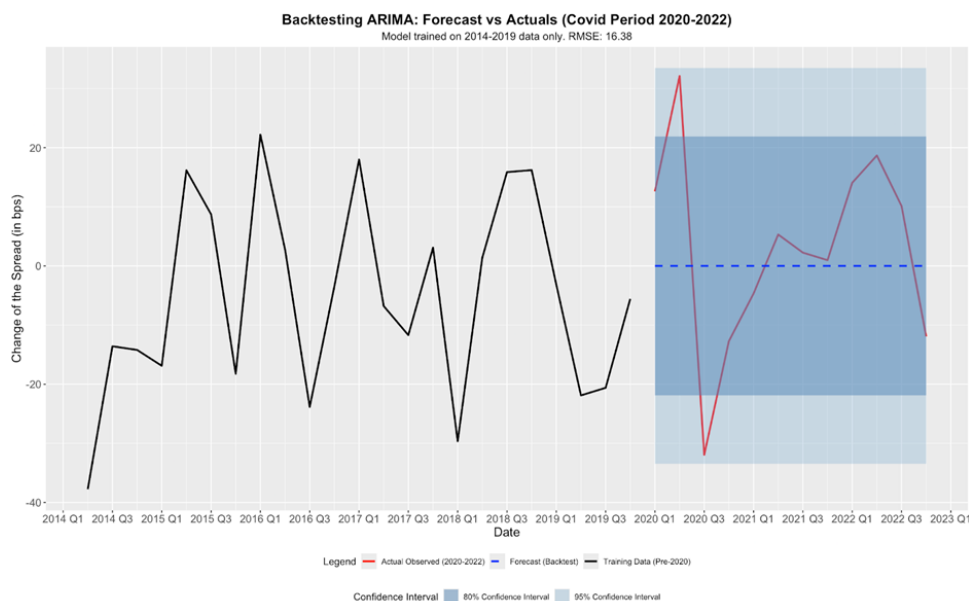


Figure 19: Backtesting ARIMA: Forecast vs. Actuals for the Covid-19 Period (2020–2022)

Predictive performance (Backtesting): To check how well the ARIMA model forecasts with our spread government bond yield, we ran a backtest on the 2020-2022 period. We hid the

covid crisis data and acted as if we did not know about the shock yet to see what the model would predict. For the training period between 2014 and 2019, we found an ARIMA(0,0,0) model. This means that the changes in the spread during those years were basically white noise, with no clear patterns or trends that the model could identify. Because of this, the safest statistical prediction for the future is just the mean of the past data, which is zero in our case because we are using first differences.

We obtained a MAE of 13.1139 and a RMSE of 16.3794 for this backtest. While the forecast is just a flat line at zero, it is useful because it shows us how far the real data moved away from the historical average during the crisis. As seen in the plot, the actual spread was very volatile with big jumps while the model stayed at zero. These error results are high, but it makes sense because a basic model cannot guess a sudden global event like the pandemic. This step helps us understand the limits of simple models when facing extreme economic shocks.

## 4 Multivariate forecast

### 4.1 Var-X and VECM-X models theoretical explanation

The Vector Autoregressive model with exogenous variables (VAR-X) and the Vector Error Correction Model with exogenous variables (VECM-X) represent advanced econometric frameworks specifically engineered for high-dimensional multivariate forecasting. These models are structured to distinguish between the dynamic interdependencies of endogenous variables and exogenous variables. The former variables are internal to the system and exhibit mutual feedback. The latter act as independent external drivers. By incorporating exogenous factors, the models account for significant external shocks that impact the system without being determined by it, thereby enhancing the precision of the estimated coefficients. While both models analyze these complex interactions, they differ in their handling of non-stationary data: the VAR-X typically requires data to be differenced to achieve stationarity, whereas the VECM-X exploits the cointegrating relationships between variables in their levels to capture long-run equilibrium dynamics alongside short-run adjustments.

### 4.2 Autoregressive Model with Exogenous Variables (VAR-X)

The VAR-X model build on the vector Autoregressive (VAR) model, which is a multivariate time series framework where each variable is modeled as a linear function of its own past values and the past values of all other variables in the system. We consider all 8 time series identified in the theoretical part as potential endogenous or exogenous variables.

When the system is augmented with exogenous variables ( $X$ ), it becomes a VAR-X model. This allows us to account for external shocks affect the endogenous system but are not themselves influenced by the internal dynamics of that system. In our case, since all primary time series are non-stationary ( $I(1)$  or  $I(2)$ ), we need to ensure stationarity and avoid spurious regressions. This step is managed by taking the first difference of the time series until the p-value of their adf test is  $\leq 0.05$ . In our case, some time series had to be differentiated twice to ensure stationarity. This transform captures the short-run relationships and the immediate velocity of change between variables.

The VAR-X(p) model can be represented mathematically a vector of  $k$  endogenous variables and  $m$  exogenous variables as such:

$$\Delta y_t = c + \sum_{i=1}^p \Phi_i \Delta y_{t-i} + \Theta x_t + \epsilon_t$$

In this structure, the `VAR()` function utilizes Ordinary Least Squares (OLS) to estimate the coefficients  $\phi$  and  $\theta$  by minimizing the sum of squared residuals for each equation independently. The exogenous component  $\Theta x_t$  acts as a fixed regressor that shifts the intercept of the system at each time step and represents the unilateral impact of the exogenous variable on the dependent variable.

### 4.3 Data Preparation and Implementation Workflow

The validity of the VAR-X model depends strictly on the statistical properties of the input data. The first necessary step is the removal of the outliers, using the Interquartile Range (IQR) method, where data points falling outside the  $1.5 \times IQR$  boundary are mitigated to prevent the OLS estimator from being biased by extreme volatility. This step reveals that while sovereign bond yield spreads and inflation spreads were relatively stable, variables such as GDP growth, Unemployment Rates, and Trade Balances exhibit significant outliers concentrated in 2020 Q2 and Q3. Outliers are replaced with the mean of the non-outlier values. For instance, the GDP spread has required three modifications, with extreme quarterly swings (e.g., 17.39% and -18.6%) being capped to a median-adjusted level of -4.76%. By doing this, we reduce the variance of the distribution. Following this, stationarity is verified using the Augmented Dickey-Fuller (ADF) test. If the null hypothesis of a unit root can't be rejected, we have to differentiate the time series until stationarity is achieved (p-values  $\leq 0.05$ ).

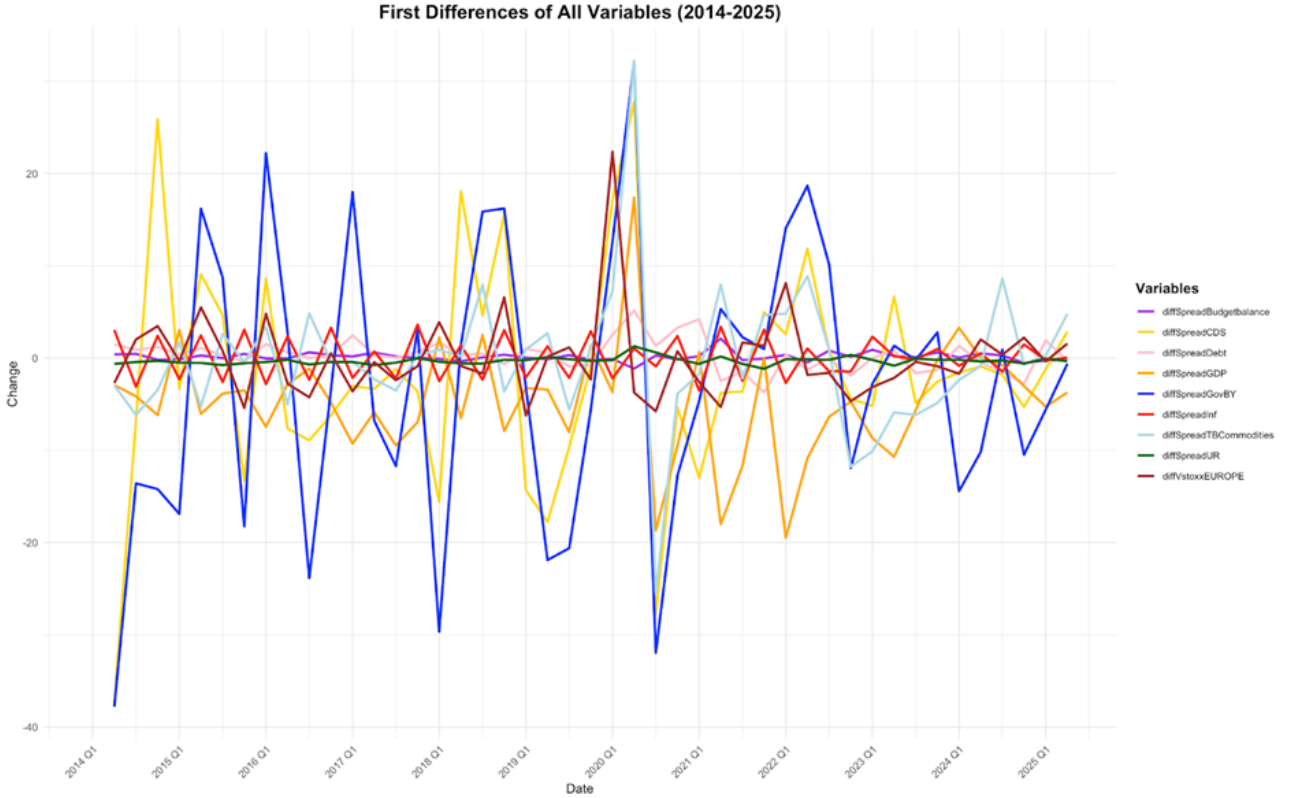


Figure 20: First differences of the time series

Once our dataset is fully stationary, we search for the optimal lag length with the `VARselect()` function. Both the Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) agree on an optimal lag of  $p = 4$ .

## 4.4 Granger tests and selection of causal variables

To identify the dynamic interdependencies within the system, bidirectional Granger causality tests were conducted in order to determine the direction of influence between the yield spread and its potential drivers. Under this framework, variables exhibiting a mutual feedback loop are classified as endogenous, while those that unilaterally affect the yield spread without being influenced by it are treated as exogenous.

The tests are computed with the `grangertest()` function and results show that the CDS spread has emerged as the most robust predictor of the government bond yield spread, with highly significant p-values at lags 1 (0.005) and 2 (0.030), justifying its role as a primary endogenous variable, even though the yield spread does not have a powerful impact on CDS. The Unemployment Rate and GDP growth are retained as exogenous variables due to marginal p-values (0.058 and 0.079, respectively) and as they should influence the yield spread theoretically. In contrast, inflation, debt, and budget balance are excluded due to a lack of influence power in both direction of granger causality. Trade Balance and Vstox are also omitted as the Granger test shows none of their past lags has any influence on the yield spread, even though they are strongly influenced by the spread. The fact that yields influence trade and market volatility highlights its role as a leading indicator in the system. Only taking the most significant variables allows us to construct a mathematically meaningful model and reduce residuals. Thus, reducing the size of the endogenous and exogenous blocks helps us better capture and forecast yield spread movements.

Following this selection, the AIC identified an optimal lag of  $p = 3$  for the endogenous system. We fixed the lag of the exogenous variables to 2, as on this lag both GDP and Unemployment have the strongest effect on the yield spread.

## 4.5 VAR-X model computation

The VAR-X(3,2) model was then estimated, regressing the endogenous vector  $\Delta y_t$  on three lags of itself and the exogenous vector  $x_t$ . The "deployed" matrix form for this system is:

$$\begin{bmatrix} \Delta spreadBY_t \\ \Delta spreadCDS_t \end{bmatrix} = \begin{bmatrix} c_1 \\ c_2 \end{bmatrix} + \sum_{i=1}^3 \Phi_i \begin{bmatrix} \Delta spreadBY_{t-i} \\ \Delta spreadCDS_{t-i} \end{bmatrix} + \begin{bmatrix} \theta_{11} & \theta_{12} \\ \theta_{21} & \theta_{22} \end{bmatrix} \begin{bmatrix} \Delta spreadg_{t-2} \\ \Delta spreadU_{t-2} \end{bmatrix} + \begin{bmatrix} \epsilon_{1,t} \\ \epsilon_{2,t} \end{bmatrix}$$

By plotting the fitted vs. actual values, we can visually assess the model's in-sample track record.

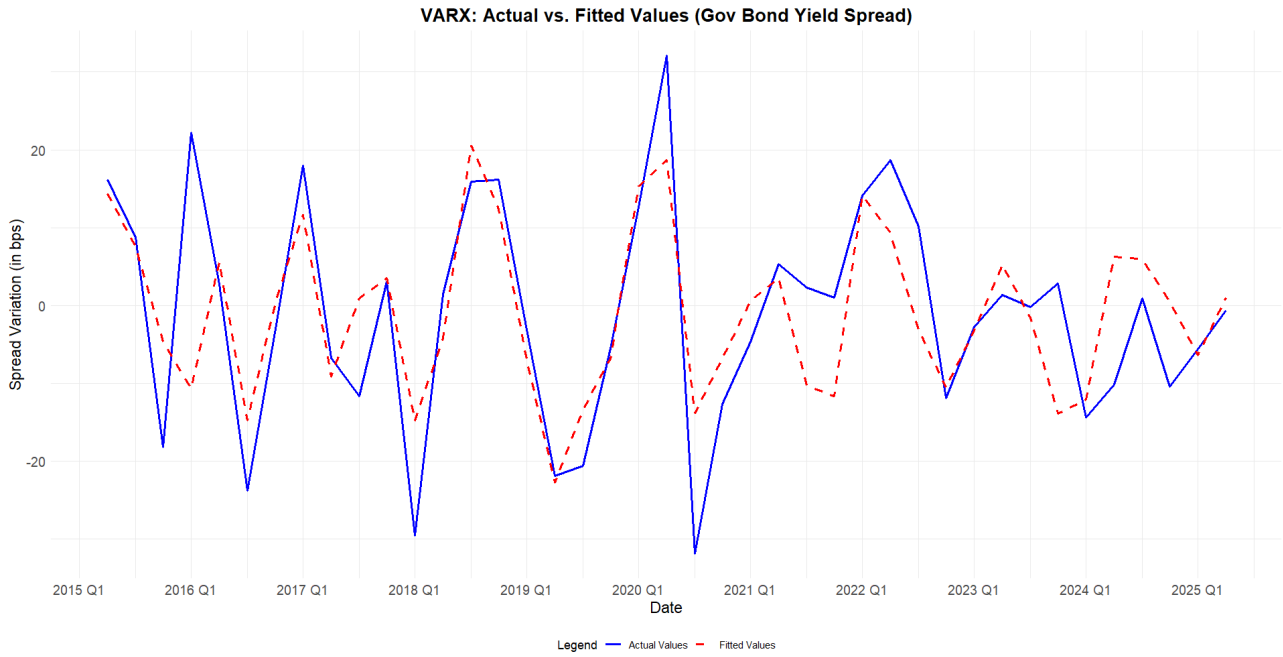


Figure 21: Actual vs values fitted by VAR-X(3)

## 4.6 Forecasting of the VAR(X) fitted values

With the model parameters finalized, we projected the system's trajectory through a forecast from 2025 Q3 to 2028 Q3. This long-term forecast relies on the estimated coefficients to predict how the yield and CDS spreads will evolve, assuming the exogenous variables continue their current trends. Mathematically, the forecast is generated recursively where each step uses the previously predicted values of the endogenous variables to determine the next period's state.

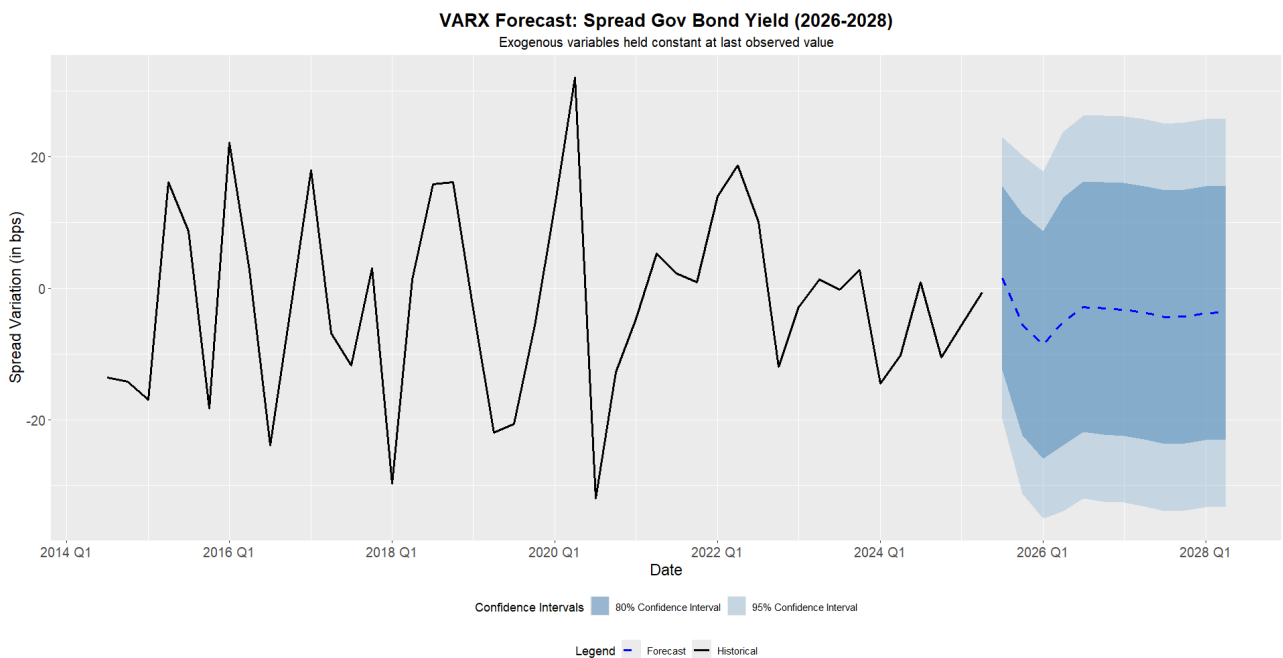


Figure 22: Forecast of the ES-DE yield spread (Q3 2025-Q3 2028)

## 4.7 Bootstrapped IRF

To understand better the structural relationship between our two endogenous variables, we use the Impulse Response Function (IRF) method which illustrates the dynamic transmission of credit risk shocks to the sovereign yield spread, which we bootstrap to construct confidence intervals around its function.

With IRF, we analyze the impact of a one-standard-deviation shock to the Credit Default Swap (CDS) spread on the Government Bond Yield spread over an eight-quarter horizon and we employ the bootstrap method to generate 95% confidence intervals.

The IRF plot reveals a highly dynamic and immediate transmission mechanism. Upon the occurrence of a shock in the CDS spread at  $t = 1$ , the government yield spread exhibits a sharp, statistically significant positive spike in the second quarter ( $t = 2$ ). This suggests that a perceived increase in credit default risk leads to an almost immediate elevation in sovereign yields as investors demand a higher risk premium. Mathematically, this response captures the  $\Phi$  coefficients of the VAR-X system in motion.

Following the initial peak, the system undergoes a period of significant volatility where it dips into negative territory between the 3rd and 5th lag. Then, from the 6th quarter onward, the response flattens and returns toward the zero line. This indicates that the shock is transitory; because the model is estimated on stationary data, the effect of the shock eventually dissipates, and the system stabilizes.

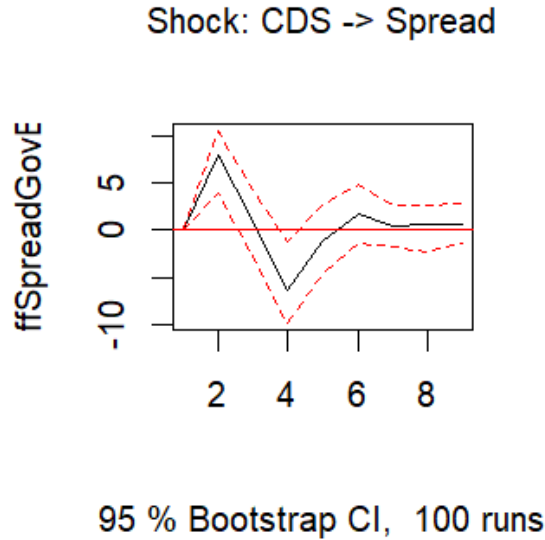


Figure 23: Analysis of the effect of CDS on the Yield Spread

## 4.8 Diagnosis of the VAR(X) errors

To evaluate the quality of our VAR-X model, we conduct the same series of diagnostic tests as before for the ARIMA on the residuals to ensure that the requirements for valid statistical inference are met. The Shapiry-Wilk (SW) test provides a p-value of 0.053, which barely confirms normality. Furthermore, the Breusch-Godfrey (BG) and Breusch-Pagan (BP) tests return p-values above 0.05 as well, confirming the residuals are white noise and the variance is

constant. Based on these results, we can believe the model is well-specified and the estimated coefficients are unbiased, with some caution regarding normality.

We also compute the error metrics MAE and RMSE, returning 6.95 and 9.65 respectively. We will use these values as a benchmark to compare with the values obtained on the VECM-X.

## 4.9 Pre- and post covid evaluation and diagnosis

To conclude the VAR-X analysis, we conduct a structural stability check by partitioning the data into two distinct subsets: the pre-covid period and the post-covid period. This comparative diagnostic analysis is essential for determining if the pandemic induced a fundamental shift in the short-run transmission of shocks or altered the predictive reliability of the model.

The diagnostic results for the subsets reveal significant differences in residual behavior, particularly regarding normality. In the pre-covid period, the Shapiro-Wilk test yields a p-value of 0.004, which means a rejection of the null hypothesis of normality. The SW test on the post-covid period returns a p-value of 0.22, thus we can't reject  $H_0$ .

Across both subsets, the model maintains good quality regarding autocorrelation and heteroscedasticity; the BG p-values (0.47 pre- and 0.92 post-covid) and BP p-values (0.28 pre- and 0.29 post-covid) confirm that the lags remain sufficient and the error variance remains stable across different economic regimes.

However, we do observe a degradation in predictive precision in the post-covid values. The MAE rose from 6.78 to 7.09 before-after covid. On the other hand, the RMSE actually decreased from 10.09 to 9.26 in the later period.

This divergence suggests that while the average error increased slightly due to persistent volatility, the frequency or magnitude of extreme values was actually higher in the pre-covid period.

## 4.10 Vector Error Correction Model with exogenous variables (VECM-X)

The VECM-X is a restricted VAR designed for variables that are non-stationary in levels but share a common long-run equilibrium, a property known as cointegration. Unlike the VAR-X, which loses long-term information through differencing, the VECM-X retains the levels of the variables within an "error correction" term. This model is theoretically superior when variables are  $I(1)$  and cointegrated because it allows the system to deviate in the short run while ensuring it returns to a stable relationship in the long run. The exogenous variables ( $X$ ) are incorporated to account for external factors that may influence the speed of this correction or shift the equilibrium itself.

Mathematically, the VECM-X with the levels reuses the VAR into a system of differences and a long-run component. The model is defined as:

$$\Delta y_t = \alpha \beta' y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Psi x_t + \epsilon_t$$

where  $\beta' y_{t-1}$  is the long-run equilibrium error from the previous period. The  $\alpha$  vector represents the speed of adjustment, dictating how much of the "error" is corrected in the current period to bring the variables back toward their cointegrated path.

## 4.11 Data preparation

The VECM-X workflow operates on the variables in their original levels after outlier removal. The first critical step is the Johansen test, which determines the cointegration rank ( $r$ ); if  $r > 0$ , a VECM is statistically justified. The model is estimated using the `cajorls` function, which applies OLS to the VECM equations once the cointegrating vectors ( $\beta$ ) have been identified.

## 4.12 VECM-X model computation

The Johansen test confirms a rank of  $r = 1$ , identifying a single stable long-run equilibrium. As we know the number of cointegration relationships, we can plot the VECM-X.

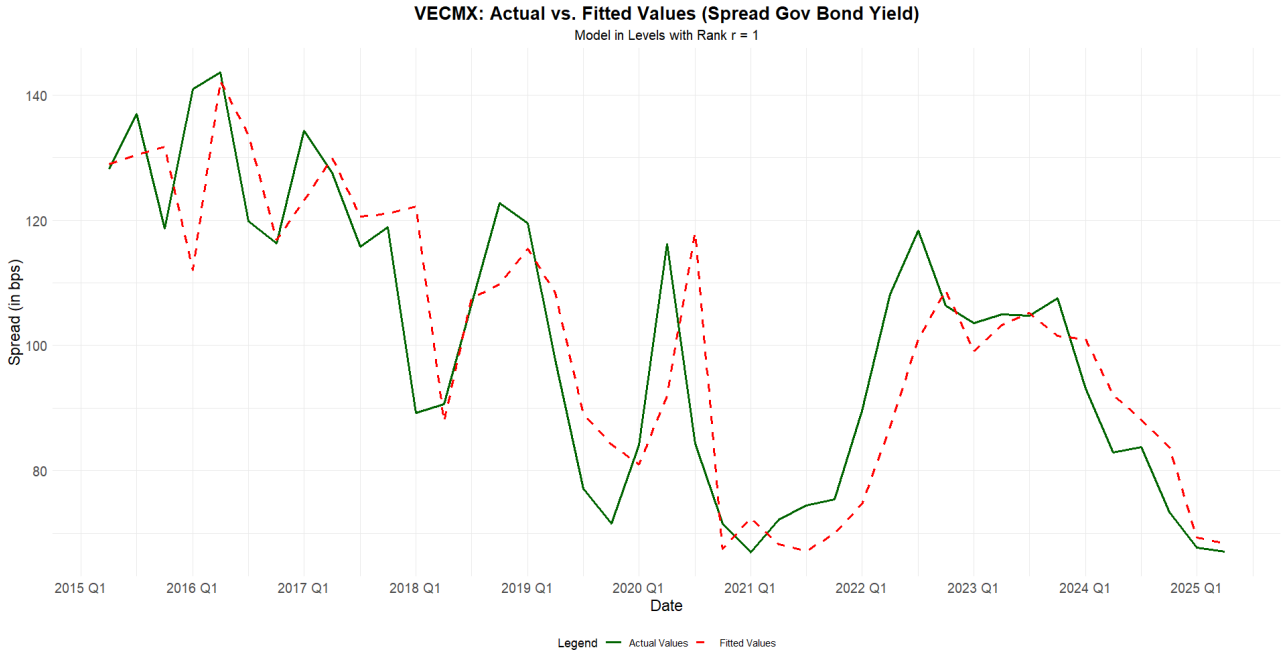


Figure 24: Actual vs Fitted values by VECM(X)

The normalized cointegrating vector ( $\beta$ ) defines this relationship as:

$$SpreadGovBY_t = 2.398 \cdot SpreadCDS_t - 259.31$$

This indicates that a 1-unit increase in the CDS spread is associated with a 2.4-unit increase in the yield spread in the long run.

The `cajorls` estimation provided a statistically significant speed of adjustment ( $\alpha$ ) for the CDS equation of 0.2854 (p-value: 0.0347), meaning 28.5% of any deviation from the long run equilibrium is corrected each quarter by CDS.

## 4.13 Diagnosis of the VECM(X) errors

The robustness of the VECM-X is confirmed through a series of diagnostic tests performed on the model's residuals. The Shapiro-Wilk test yields a p-value of 0.09, which, being above the 0.05 threshold, suggests that the residuals are normally distributed. This normality is vital for the validity of subsequent statistical inferences. Furthermore, the Breusch-Godfrey test results in a p-value of 0.99, indicating a near-total absence of autocorrelation and confirming that the four-lag structure in levels adequately captures the system's dependencies. Finally, the

Breusch-Pagan test for heteroscedasticity returns a p-value of 0.28, verifying that the residual variance remains constant over time.

The forecasting performance of the VECM-X in levels is evaluated using standard error metrics. The MAE for the model in levels is 8.91, while the RMSE is 12.38.

#### 4.14 Forecasting of the VECM(X) fitted values

The VECM-X model was utilized to generate a forecast extending from 2025 Q3 to 2028 Q3 using the predict function. Unlike the VAR-X, which focuses on short-term velocity, the VECM-X forecast is anchored by the identified cointegrating relationship. The resulting projection indicates a continuous and steady decline in the government bond yield spread over the three-year horizon.

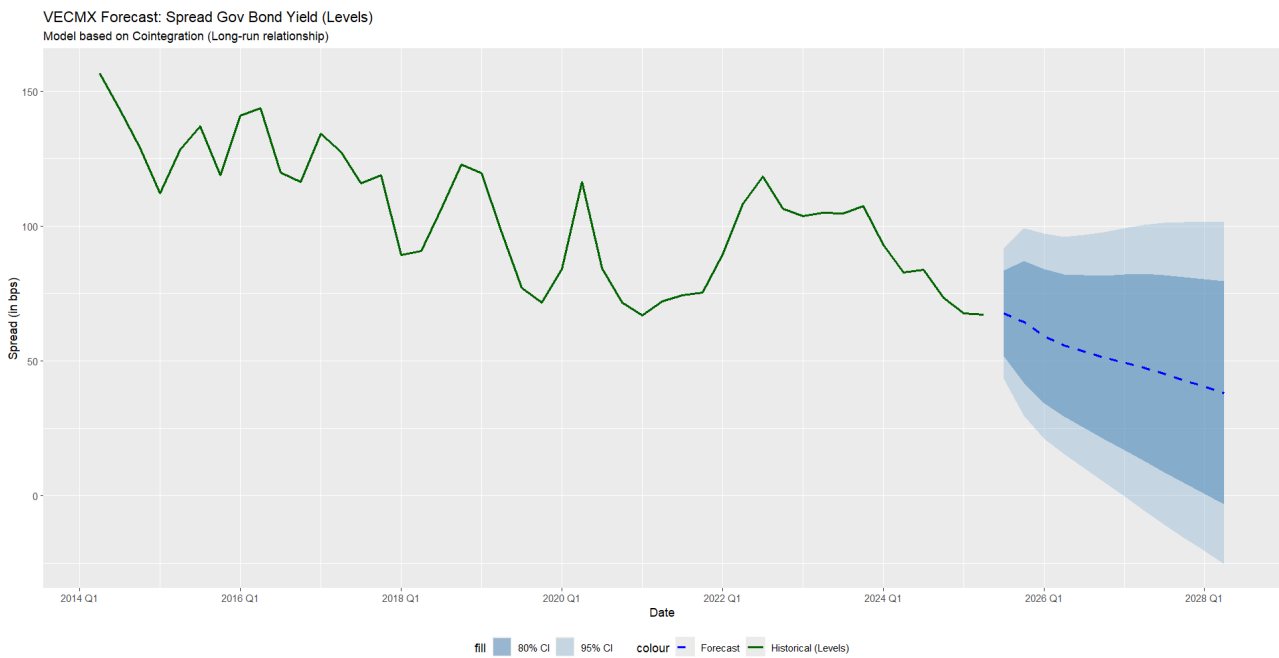


Figure 25: Forecasting result with VECM(X) model

#### 4.15 Pre- and post-covid evaluation

A comparative sub-sample analysis was conducted for the VECM-X to assess its structural stability across the profound economic shifts of the last decade. During pre-covid, the model exhibited a MAE of 9.2 and a RMSE of 12.77. In the post-covid period, the predictive accuracy improved slightly, with the MAE falling to 8.67 and the RMSE reaching 12.04. This marginal reduction in error metrics suggests that the VECM-X's reliance on long-run cointegrating ties provides a robust anchor that remains effective even during periods of high volatility. While the COVID-19 pandemic introduced severe short-term shocks, the stability of these sub-sample results indicates that the fundamental equilibrium between credit risk and government bond yield spreads remained structurally sound, allowing the model to maintain, and even slightly enhance, its precision in the post-pandemic landscape.

#### 4.16 Future outlook and forecast

The final projection generated by the VECM-X indicates a continuous and steady decline in the yield spread through Q3 2028. This forecast is driven by the model's error-correction

mechanism, which actively pulls the system back toward its historical baseline at a significant speed of adjustment ( $\alpha = 0.285$ ).

## 5 Innovation

### 5.1 Introduction theory to the Kalman filter based model

The Kalman filter is a very useful tool for modeling and predicting dynamic systems, especially when we have to deal with hidden variables or noisy data. It works within a state-space framework, which is an organized way to handle time series by splitting the system into two parts: what we can actually see and what stays hidden. This structure allows the model to understand complex dynamics while also accounting for random noise in the observations and the system.

A state-space model is built with two main equations.

First, there is the state equation:

$$X_t = F \cdot X_{t-1} + \epsilon_t$$

This equation explains how the hidden state vector,  $X_t$ , changes from one period to the next. These states represent the "behind the scenes" dynamics that we can not observe directly. The matrix  $F$  (state transition matrix) shows how the state moves from the previous step ( $X_{t-1}$ ) to the current one, and  $\epsilon_t$  is the process noise, which represents random shocks or uncertainty in how the system evolves.

Second, there is the observation equation:

$$y_t = H \cdot X_t + \eta_t$$

This equation connects the data we actually measure ( $y_t$ , like the spread bond yield) to the hidden states ( $X_t$ ). The matrix  $H$  (observation matrix) is what maps these hidden states to the real-world observations. Finally,  $\eta_t$  is the observation noise (with its variance =  $R$ ), which represents measurement errors or the inherent inaccuracy of your data quality. By using these two equations together, the filter can estimate the hidden states more accurately every time new data  $y_t$  becomes available.

Another important part of the theory is the Kalman Gain, which we represent as  $K_t$ . This is basically the weight that the model uses to decide how much it should believe the new data  $y_t$  it just received compared to its own internal guess. When the model makes a prediction, it compares that prediction to the real observation. The difference between the real data and what the model expected is called the innovation. The Kalman Gain then determines how much of this innovation should be added to the prediction to get the final, corrected estimate. A high Kalman gain is chosen when the uncertainty of the model's state ( $P$ ) outweighs the observation noise ( $R$ ), indicating that the incoming data is highly reliable and that thus the model gives more importance to this new observation. We have this equation:  $K_t = \frac{P_{t|t-1}}{P_{t|t-1} + R}$

Consequently, the filter aggressively adjusts its estimates to closely track the latest observation, as it believes the innovation represents a true shift in trajectory rather than just a random noise. On the other hand, If the observation noise ( $R$ ) is very high in contrast with the uncertainty of the model's state ( $P$ ), the model thinks the new data is not very reliable, so  $K_t$  stays small and the model trusts its own prediction more.

This configuration ensures the system remains extremely reactive to real-time changes in the data.

We can see this in this equation:

$$X_t = \hat{X}_t + K_t(y_t - H\hat{X}_t)$$

In this formula,  $\hat{X}_t$  is the prediction of the state the model made before seeing the new data, and the term  $(y_t - H\hat{X}_t)$  is the innovation.

## 5.2 Our Kalman Filter: Model Setup

To initiate our innovation phase, we constructed a dedicated dataset, `df_kalman`. This dataset integrates our target variable, `diffSpreadGovBY`, with a trio of macroeconomic predictors: `diffSpreadCDS`, `diffSpreadUR`, and `diff2SpreadGDP`. A fundamental step in our pre-processing is the treatment of outliers using the Interquartile Range (IQR) method across all four variables. This ensures that extreme, non-representative shocks do not distort the Kalman filter’s ability to accurately estimate the underlying hidden states, as the algorithm is highly sensitive to noise that can be mistaken for structural shifts.

For the specification of the state matrix  $X_t$ , which contains all hidden states, we utilized the results of the multivariate bidirectional Granger causality tests. We prioritized variables demonstrating the strongest exogeneity in the direction of causing the target spread. Consequently, our hidden state vector includes the sensitivities for `diffSpreadCDS`, `diffSpreadUR` and `diff2SpreadGDP`. These variables proved to be the most relevant predictors, collectively capturing market risk, labor market dynamics and economic growth.

The structure of our state-space model was built by combining a multivariate dynamic regression with a seasonal component using the `dlm` package in R, which is specifically designed for building dynamic linear models within a state-space framework. We are configured to track seven distinct hidden states: the moving intercept, the time-varying Beta coefficients for the CDS, GDP, and Unemployment Rate, and three states for quarterly seasonality. By using `dlmModSeas` with a frequency of 4, the model captures the regular annual cycles in `diffSpreadGovBY`, effectively isolating these seasonal patterns from the structural impact of our three exogenous predictors.

To account for uncertainty in the system, we define the process noise covariance matrix  $Q$  ( $W$ ) and the observation noise variance  $R$  ( $V$ ). For the state transition described by the equation  $X_t = F \cdots X_{t-1} + \epsilon_t$ , we construct  $Q$  as a diagonal matrix to represent the noise affecting our seven hidden states. This diagonal structure assumes that the intercept, the three economic coefficients, and the seasonal components evolve independently over time. We specifically assign variances to these elements to allow them to be dynamic, which is essential for capturing how the relationship between the predictors and the spread changes during volatile periods. In the observation equation,  $y_t = H \cdots X_t + \eta_t$ ,  $R$  ( $V$ ) represents the observation noise. All noise components are parameterized using exponential functions during optimization to ensure that the variances always stay positive. Finally, we set robust initial conditions with a starting mean `m0` of zero and an initial covariance `C.0` of 10. This high initial uncertainty allows the model to adapt its estimates quickly to the data at the start of the sample.

To find the optimal parameters, we used the `dlmMLE` function to perform maximum likelihood estimation. This function adjusts six different variance parameters specifically for  $R$  ( $V$ ), the intercept, the three beta coefficients, and the seasonal component to minimize the prediction error. Once these parameters are optimized, we rebuild the final model and apply the `dlmFilter`

function. This function processes the data recursively, updating its estimates every time a new observation becomes available by calculating the Kalman Gain. This high-dimensional setup allows the model to reach a steady state where the gain is optimized, providing a robust foundation for our 12-quarter forecast.

### 5.3 Our Kalman Filter: Algorithm

Mathematically, the algorithm follows a prediction and update loop. In the prediction step, it forecasts the next state and its uncertainty based on the previous estimate:

$$\hat{x}_{t|t-1} = F\hat{x}_{t-1|t-1}$$

This step propagates the information contained in the past state through the state transition matrix. Once the state is predicted, its associated uncertainty (P) must also be updated to reflect the accumulation of process noise (Q) and model uncertainty:

$$P_{t|t-1} = FP_{t-1|t-1}F^\top + Q$$

The model then predicts the observation:

$$\hat{y}_{t|t-1} = H\hat{x}_{t|t-1}$$

When the new data point arrives, the update step begins. The model first computes the innovation, which measures the discrepancy between the observed value and the predicted observation:

$$v_t = y_t - \hat{y}_{t|t-1}$$

To assess how reliable this innovation is, the model then calculates the innovation covariance, which combines the uncertainty of the predicted state with the observation noise:

$$S_t = HP_{t|t-1}H^\top + R$$

Based on this uncertainty, the Kalman gain is computed. It determines how much weight should be given to the new observation relative to the model's prior prediction:

$$K_t = P_{t|t-1}H^\top S_t^{-1}$$

Using this gain, the state estimate is updated by correcting the predicted state with the innovation:

$$\hat{x}_{t|t} = \hat{x}_{t|t-1} + K_tv_t$$

At the same time, the state covariance matrix is updated to reflect the reduction in uncertainty after incorporating the new observation. Because the model has learned from the data at time  $t$ , the uncertainty around the state estimate decreases:

$$P_{t|t} = P_{t|t-1} - K_tHP_{t|t-1}$$

The updated estimate  $\hat{x}_{t|t}$  represents the best possible estimate of the hidden state at time  $t$ , as it combines both the model's prediction and the information contained in the latest observation. Once this update step is completed, the algorithm moves forward to the next time period,  $t+1$ . The updated quantities  $\hat{x}_{t|t}$  and  $P_{t|t}$  then become the starting point for the next prediction step, and the prediction-update cycle continues recursively over time. The Kalman filter repeats these prediction and update steps for every single time step in the data. Each new iteration uses the results from the previous step as its starting point, which makes the model very efficient and allows it to adapt quickly as soon as new data is available. Once we have processed all

historical observations, the filter can be used to forecast the future. To do this, we run only the prediction step repeatedly without the update part, using the final state estimate and the model parameters:  $F$ ,  $Q$ ,  $H$  and  $R$  found during the filtering process. This algorithm is highly flexible because it accounts for both measurement and process noise while working efficiently. Its ability to handle noisy information and estimate hidden states makes it a very important tool for modeling dynamic systems. In our project, it predicts spread variations by combining several economic indicators into a single framework, balancing the actual observed data with the system's internal dynamics to provide reliable and accurate forecasts.

## 5.4 Our Kalman Filter: Forecasting

For the forecasting component, we extend our timeline by 12 quarters to cover a 3-year horizon. Since our Kalman model is a dynamic regression, it requires future values for the predictors `diffSpreadCDS`, `diffSpreadUR`, and `diff2SpreadGDP` to estimate the future bond spread. We obtain these values by using the `auto.arima` function to find the most suitable stochastic processes for each variable. For the target variable, we create a  $y$  full vector where all future dates are marked as NA. This technical setup is essential because when the `dlmFilter` function encounters an NA, it automatically switches from its standard updating mode to a pure prediction mode. This means it stops adjusting its hidden states based on real-time observations and starts projecting them into the future using only the underlying system equations and the predicted paths of our regressors.

The heart of this model is a recursive cycle that adapts based on whether data is available. In the historical phase where we have actual values for the bond spread, the filter uses each new observation to correct its coefficients through the Kalman Gain, learning how the spread reacts to external shocks. However, during the forecast period where the spread is unknown, the model relies on the state it reached at the last known data point. The equation used to generate the forecast remains:

$$f_t = \alpha_{t|t-1} + (\beta_{CDS,t|t-1} \cdot CDS_t) + (\beta_{UR,t|t-1} \cdot UR_t) + (\beta_{GDP,t|t-1} \cdot GDP_t) + S_{t|t-1}$$

In this forecasting phase, the filter applies the last known projected coefficients for the intercept, the seasonal component, and the sensitivities to the future values of the predictors. This allows the model to simulate how the spread will move in response to future economic changes even without real spread data to learn from. This mechanism is supported by our Granger causality tests, which proved that past movements in CDS and unemployment are reliable indicators of future spread behavior.

To generate the future inputs for this equation, we identified specific ARIMA models for each predictor: `diffSpreadCDS` follows an ARIMA(4, 0, 0), `diffSpreadUR` follows an ARIMA(0, 0, 0), and `diff2SpreadGDP` follows an ARIMA(2, 0, 0). The latter indicates that the GDP variable has no clear trend for the future, which is a cautious assumption for a risk variable. By integrating these specific paths into a re-defined structure using `dlmModReg` and `dlmModSeas`, we ensure the filter maintains its quarterly seasonal cycles throughout the 3-year extension. Reusing the optimized variances from the maximum likelihood estimation step ensures the final projection is not just a flat trend, but a dynamic forecast that follows the predicted paths of the exogenous variables while respecting the historical quarterly patterns identified during the training phase.

## 5.5 Kalman Filter results

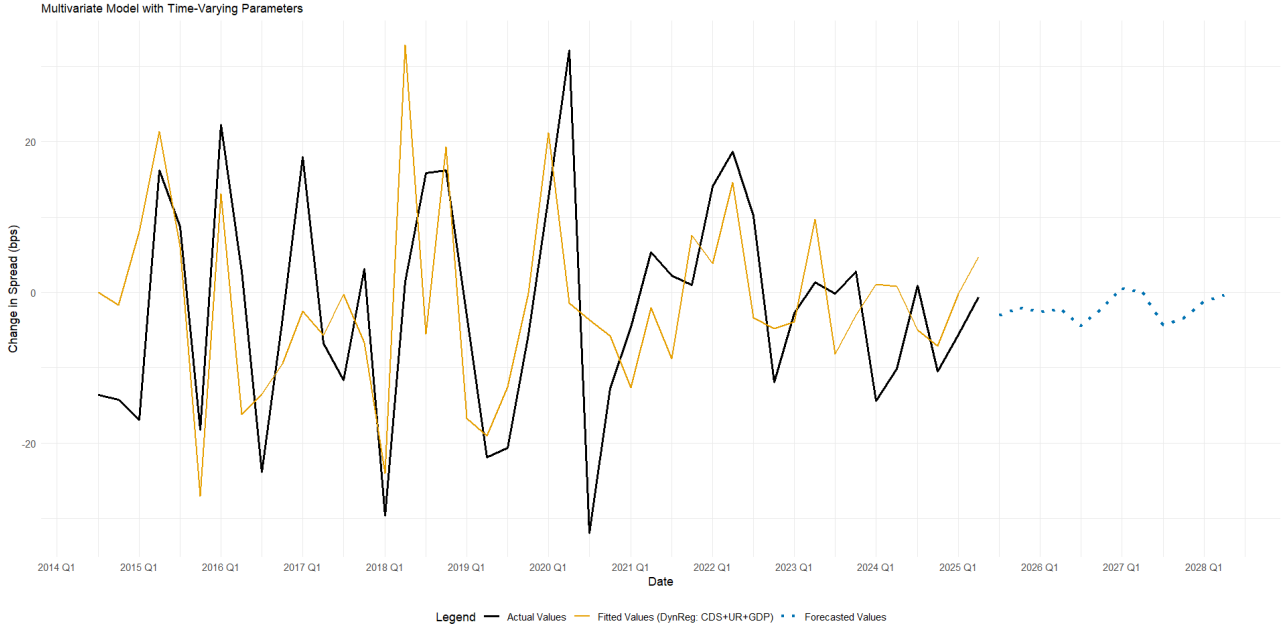


Figure 26: Kalman Filter: Dynamic Regression

The plot showcases the actual, fitted, and forecasted change in spread over time, with the x-axis representing the dates from 2014 to 2028 and the y-axis displaying the change in basis points of the `SpreadGovBY`. The solid black line represents the actual observed values for `diffSpreadGovBY`, which shows a high level of volatility with significant peaks and troughs. The yellow line represents the fitted values generated by our Kalman filter during the historical period. This fitted line stays closely aligned with the actual data, demonstrating that the model is effective at capturing the movements of the spread while smoothing out some of the most extreme noise to identify the underlying states.

The blue dotted line illustrates the forecasted values for the change in spread over a three-year horizon. We can clearly see that the forecast does not follow a straight horizontal path but instead forms a series of waves. This is a very robust result because it shows that the Kalman filter has successfully separated the structural trend from the seasonal cycle. The Kalman filter remembers that the government bond yield spread between Spain and Germany typically follows a quarterly rhythm. These waves represent the regular seasonal tensions that occur every year, such as specific market behaviors in the first or last quarters. We can also see that forecasted values remain slightly below or on the zero line, mostly in a  $[-5, 0]$  bps range. This indicates that the model predicts a gradual but fluctuating narrowing of the spread between Spain and Germany over the next three years. Such a trend suggests a slow convergence toward German yield levels, which represents a positive economic signal for Spain.

In summary, the presence of these waves in the forecast makes the model much more sophisticated than a basic linear projection. Instead of predicting a relatively flat bond yield market, the model suggests that the spread will remain stable in terms of its general level while continuing to fluctuate according to its historical seasonal patterns. This combination of structural stability from the ARIMA part and cyclical movement from the Kalman seasonality provides a more realistic view of how the spread might behave. It confirms that the model is sensitive to the internal timing of the data rather than just the external predictors.

After that, we check if the model is reliable by looking at the residuals, which are the differences between the real data and our fitted values with our Kalman filter model. We calculate

the MAE and RMSE to measure the size of the errors, and we also run several statistical tests. We use the Shapiro-Wilk test for normality, the Breusch-Pagan test for heteroscedasticity, and the Breusch-Godfrey test to check for autocorrelation of the residuals. We also compare the performance before and after the COVID crisis to see if the model adapted well to the high volatility of the pandemic.

## 5.6 Model evaluation

| Metric                                     | Model (Global) | Pre-COVID | Post-COVID |
|--------------------------------------------|----------------|-----------|------------|
| Shapiro-Wilk Normality (p-value)           | 0.5999         | 0.6960    | 0.3059     |
| Breusch-Pagan Heteroscedasticity (p-value) | 0.8266         | 0.4695    | 0.6873     |
| Breusch-Godfrey Autocorrelation (p-value)  | 0.8878         | 0.9511    | 0.4207     |
| MAE                                        | 10.4851        | 11.2019   | 9.7683     |
| RMSE                                       | 12.9658        | 13.6240   | 12.2724    |

Table 2: Model evaluation metrics across different periods

The table summarizes the evaluation metrics for the model across three scenarios: the entire model (Global), the pre-COVID period, and the post-COVID period. These metrics include the Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Breusch-Godfrey test p-values, Shapiro-Wilk test p-values, and Breusch-Pagan test p-values, which provide a detailed assessment of the model’s performance and the residuals’ statistical properties.

The diagnostic results demonstrate strong model validity. The Shapiro-Wilk p-value of 0.5999 (Global) indicates that the residuals follow a normal distribution. High p-values for the Breusch-Pagan (0.8266) and Breusch-Godfrey (0.8878) tests confirm that the model is free from heteroscedasticity and autocorrelation issues. The model remains highly effective in the post-COVID period, as evidenced by an MAE of 9.7683 and RMSE of 12.2724.

The MAE, which measures the average magnitude of the residuals, is 10.4851 for the global model. In the pre-COVID period, the MAE increases to 11.20, while in the post-COVID period it declines to 9.7683, indicating smaller average prediction errors and improved model performance after the onset of the pandemic. A similar pattern is observed for the RMSE, which places greater weight on larger errors. The RMSE equals 12.97 for the global model and rises to 13.62 in the pre-COVID period, before decreasing to 12.27 in the post-COVID period. Overall, these results suggest that the model captures the underlying data dynamics more accurately in the post-COVID period than in the pre-COVID period.

The Breusch-Godfrey test evaluates autocorrelation in the residuals up to order 4. The p-value for the entire model is 0.89, suggesting no significant autocorrelation. Pre-COVID, the p-value is 0.95, which strongly confirms autocorrelation of the residuals. Post-COVID, the p-value is 0.42, which is lower than the previous periods but still well above the 0.05 threshold, indicating that there is no autocorrelation issue in any of the periods.

The Shapiro-Wilk test assesses whether the residuals are normally distributed. For the entire model, the p-value is 0.60, suggesting that the residuals follow a normal distribution. In the pre-COVID period, the p-value is 0.70, further supporting the normality assumption. Post-COVID, the p-value decreases to 0.31. Although this value is closer to the significance level, it remains well above 0.05, meaning the residuals can be considered normally distributed in the post-pandemic period.

The Breusch-Pagan test confirms heteroscedasticity with a global p-value of 0.83, indicating very strong evidence of homoscedasticity. For the pre-COVID period, the p-value is 0.83, and for the post-COVID period, it is 0.69. Since all these p-values are significantly higher than 0.05,

we can conclude that the variance remains constant across all timeframes and heteroscedasticity is not a problem for this model.

In conclusion, the Kalman filter model demonstrates robust performance across all tested periods. Interestingly, the model shows better accuracy during the post-COVID period as reflected by the lower MAE and RMSE values. Furthermore, the model maintains its statistical validity throughout the entire timeframe, as the residuals consistently pass the tests for no autocorrelation, normality, and constant variance. These results indicate that the model is reliable and effectively captures the underlying dynamics of the data.

## 5.7 Time-Varying Sensitivity of Spread to predictors

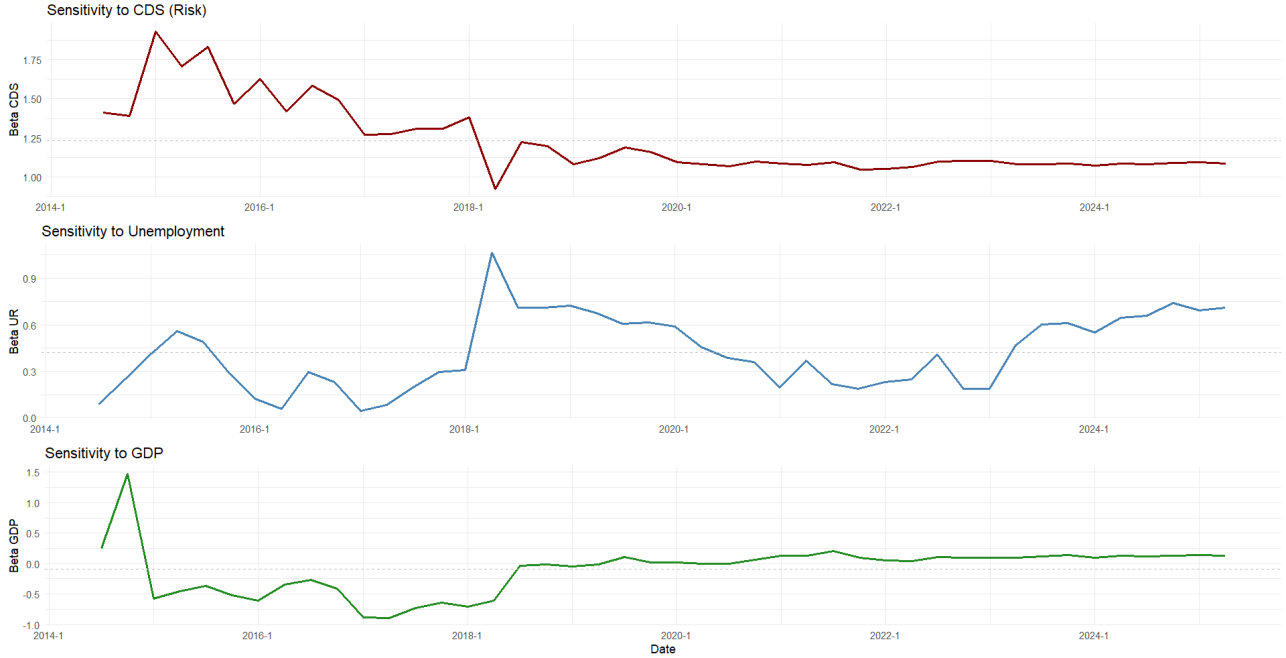


Figure 27: Yield Spread Time-sensitivity to predictors

The chart shows the time-varying sensitivity of the spread to the predictors, which is represented by the evolution of the Beta coefficient over time. At the start of the study period in 2014, the sensitivities across all three predictors exhibit high volatility and sharp fluctuations, particularly for CDS and GDP where values reach extremes above 1.5. This initial phase from 2014 to 2016 is called the burn-in or initialization period, where the Kalman filter is learning the underlying relationships from a limited number of data points, making the state estimates highly reactive to every new piece of information. During this window, the CDS coefficient suggests a dominant impact on the spread, while the GDP and Unemployment coefficients show rapid adjustments as the model seeks to distinguish signal from noise. This justifies the dynamic approach, as the filter must calibrate itself to the specific weights of all the predictors.

However, a clear transition toward stabilization occurs after 2018 for CDS and GDP, leading into a much more consistent phase for the remainder of the study. Unemployment beta shows a stark increase to a peak of 1 from Q1 and Q2 2018. Then it gradually decreases but stays volatile until 2025. From 2020 through 2025, the sensitivity to CDS remains relatively constant at a lower level around 1.1, while GDP sensitivity flattens near zero, indicating that the model has reached a steady state where the Kalman gain is optimized. Even as the sensitivity to Unemployment shows a somewhat stable increase toward the end of the period, the lack of erratic swings suggests the filter has finally identified the true underlying signal despite the

inherent noise in the data. This stabilization explains the improved predictive accuracy during the post-COVID period, as shown by lower error metrics.

## 5.8 Final comparison and conclusion

| Model               | MAE Global | RMSE Global | MAE Pre-Covid | RMSE Pre-Covid | MAE Post-Covid | RMSE Post-Covid |
|---------------------|------------|-------------|---------------|----------------|----------------|-----------------|
| ARIMA (Baseline)    | 10.1693    | 13.1898     | 11.3481       | 13.8919        | 9.1513         | 12.5520         |
| VARX (Multivariate) | 6.9511     | 9.6528      | 6.7892        | 10.0915        | 7.0909         | 9.2572          |
| Kalman Filter       | 10.0056    | 12.5232     | 10.2804       | 12.8075        | 9.7683         | 12.2724         |

Table 3: Quality Analysis

| Test Comparison    | DM Statistic | P-value | Final Result                 |
|--------------------|--------------|---------|------------------------------|
| 1. Kalman vs ARIMA | -0.38791     | 0.3501  | No significant difference    |
| 2. Kalman vs VARX  | 1.3585       | 0.909   | No significant difference    |
| 3. VARX vs ARIMA   | -2.1543      | 0.01865 | VARX is significantly better |
| 4. VARX vs Kalman  | -1.3585      | 0.09095 | No significant difference    |

Table 4: Test comparison

In conclusion, this project evaluated four different modeling approaches to predict the government bond yield spread between Spain and Germany: the ARIMA, the multivariate VARX, the multivariate VECMX, and the Kalman filter. To ensure a fair comparison, the evaluation was conducted on a strictly aligned timeframe by using the shortest residual segment available. This segment belonged to the VARX model because it loses observations due to the use of lags and double differencing on the variables in the dataframes. This alignment ensured that all performance metrics were calculated over the exact same period, from Q2 2015 to Q2 2025. It is also important to mention that the VECMX model was excluded from this final comparison because it operates on data in levels, whereas the other three models use differenced data, making their residuals mathematically incomparable for a direct battle of models.

The performance metrics show that the VARX model is the most accurate of the three across the entire period. It achieved a global Mean Absolute Error of 6.9511 and a Root Mean Squared Error of 9.6528, which are notably lower than the results for both ARIMA and the Kalman filter. While the Kalman filter provided better results than the ARIMA with a global RMSE of 12.52 compared to 13.18 it did not reach the same level of precision as the VARX model. When splitting the data between pre-COVID and post-COVID periods, the VARX model continued to lead, maintaining lower error values in both scenarios, while all models generally showed slightly better accuracy in the post-pandemic timeframe.

To determine if these differences in error were statistically significant, we performed Diebold-Mariano tests. The results clearly identify the VARX model as the superior prediction tool. The test comparing VARX to ARIMA yielded a p-value of 0.01865, meaning we can conclude that VARX is significantly better than the baseline at a 5% significance level. Interestingly, during the Diebold-Mariano test of VARX vs Kalman, the VARX model almost managed to be significantly better than the Kalman filter with a p-value of only 0.09. When comparing the Kalman filter to ARIMA or VARX, the results did not show a statistically significant difference,

with p-values of 0.35 and 0.91 respectively. This suggests that while the Kalman filter is a valid and useful model, it does not offer a significant predictive gain over the simpler multivariate approach of the VARX.

Overall, the project demonstrates that our choice of incorporating in the VAR-X our variable `diffSpreadCDS` as our endogenous variable and our two exogenous variables `diff2SpreadGDP` and `diffSpreadUR` through a multivariate framework provides the most reliable forecast. The VARX model not only produced the lowest absolute errors but was also the only model to show a statistically significant improvement over the ARIMA according to the Diebold-Mariano test. Although the Kalman filter remains highly valuable for its ability to track how the spread sensitivity changes over time, as shown in the innovation analysis plot, the VARX model stands out as the best overall choice for prediction. All models passed the necessary diagnostic tests for their residuals, including normality and constant variance, which confirms that the statistical foundations of our comparison are solid.

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